Deriving the Particle Zoo from Observer Consistency

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Abstract

Observer-Patch Holography (OPH) asks how much of the particle spectrum can be derived from one shared local input rather than from a long fitted parameter list. Building on the OPH recovery of relativity and Standard Model structure, this paper develops the particle branch from the dimensionless pixel constant $P \equiv a_{\mathrm{cell}}/\ell_P^2$. It fixes the structural massless carriers, closes a target-free W/Z lane, carries a quantitative Higgs/top branch, closes the exact PDG 2025 API running-quark sextet together with explicit exact forward Yukawas on the selected public quark frame class f_P, and gives a theorem-grade weighted-cycle neutrino family in the observed hierarchy and mixing regime. The exact non-hadron surface includes exact structural zeros, an exact frozen W/Z pair, exact same-family charged witnesses, the selected-class exact quark theorem together with its supporting exact surfaces, and exact neutrino-side readout objects. Charged-lepton absolute closure, a global classification of all quark frame classes, flavor-labeled neutrinos, and hadrons remain explicit boundary objects.

1 Introduction

Observer-Patch Holography (OPH) derives physics from observer consistency. Its starting point is that physical data are organized in local patch algebras on a screen, nearby observers must agree on overlaps, realized states obey a local maximum-entropy/refinement principle, generalized entropy is recoverable, and the realized low-energy branch is selected by a minimal admissibility criterion. From that basis one can derive, conditionally but explicitly, the Lorentzian and Einstein branches together with the realized Standard Model gauge structure

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3.$$

The Lorentzian branch, Einstein recovery, and the realized Standard Model gauge structure written above are derived in [4]. That same paper also derives the hypercharge lattice and the counting results $N_g = 3$, $N_c = 3$. This paper uses those results as input and summarizes only the downstream particle-spectrum continuation. If that derivation is correct, it should not stop at gauge structure. It should also constrain the particle spectrum.

The particle branch studied here is a reverse-engineering problem. Instead of treating masses, mixings, and Yukawa parameters as independent inputs, it asks how far one common dimensionless pixel constant P can be propagated through the whole spectrum. The working hypothesis is that one fixes one external pixel input P and then uses that same P through the bosonic, quark, charged-lepton, neutrino, and hadronic branches. The structural massless carriers, the closed W

boson/Z boson electroweak pair, the Higgs boson/top quark pair, the selected-class exact quark theorem, and the weighted-cycle neutrino theorem branch are emitted, while charged leptons, flavor-labeled neutrinos, and hadrons remain non-closed, with hadrons specifically execution-contract-frozen behind one production backend export bundle plus production systematics. This paper develops that derivation chain from the axioms and P to the reported particle outputs, while keeping the open closure boundary explicit.

1.1 Why this particle zoo, and not another one?

The shortest informal answer is that, in OPH, particles are not inserted by hand as a list of elementary ingredients. They are the stable observer-visible excitations and carrier modes that remain once three layers of structure have been fixed:

1. overlap consistency on the observer-patch network;
2. compact gauge reconstruction from the transportable edge-sector colimit once its directed-colimit and fiber-functor premises are supplied;
3. minimal admissible realization of the low-energy branch.

The first layer says what kinds of local data can be compared consistently across neighboring patches. The second says that persistent transportable sector data, once the compact paper's explicit colimit and fiber-functor premises are supplied, organize into a compact gauge structure. The third selects which of the admissible low-energy branches is physically realized. At that point one is not choosing an arbitrary particle catalog. One is reading off the carrier and matter content of the realized branch.

That is why *Recovering Relativity and Standard Model Structure from Observer Overlap Consistency* [4] is so important for this draft. That paper does the structural work that fixes the realized low-energy package

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3,$$

together with one admissible Higgs doublet and the realized chiral matter content. In plain language: OPH does not begin by guessing photons, gluons, quarks, leptons, and weak bosons and then asking how to fit them. It first derives the realized gauge-and-matter branch, and that branch determines which elementary carriers and matter families are available at low energy.

This also explains the phrase "and no others." The structural OPH claim concerns the realized low-energy branch selected by the stated admissibility conditions. Once one imposes those conditions, the realized low-energy branch is the Standard-Model branch rather than a larger light-sector package. Additional light gauge bosons, additional light Higgs multiplets, extra light chiral families, or low-energy supersymmetric partners do not appear on that realized branch because they would enlarge the admissible package that MAR selects against. In that sense the observed light particle zoo is not an arbitrary menu; it is the minimal realized one.

The individual families then emerge for different reasons. The photon, gluons, and graviton are structural carriers: once the realized electromagnetic, color, and dynamical-metric branches are present, those massless carriers follow as symmetry-protected zeros. The weak bosons arise when the electroweak gauge sector is calibrated quantitatively. Quarks and leptons arise from the realized chiral matter package together with the three-generation structure $N_g = 3$. Their family splittings are then read from the deeper overlap-transport and excitation machinery, not from a second arbitrary postulate that says "copy the family three times and assign masses by hand."

Hadrons come one layer later still: once color is realized and confined, stable hadrons are the composite readout channels of the quark/gluon sector rather than separate fundamental inputs.

The role of the pixel constant P is also important to state clearly. Here, P does not decide whether the photon exists, whether there are three colors, or whether quarks come in three generations. Those structural facts belong to the gauge-and-matter branch fixed before calibration. What P does is set the shared quantitative scale on which the realized particle content is read out numerically. So the logic of the paper is:

$$\begin{aligned}\text{observer consistency} &\to \text{realized gauge/matter branch} \to \text{particle content}\\ &\to \text{common scale } P \to \text{quantitative spectrum.}\end{aligned}$$

That is the sense in which the derivation explains why a universe with the realized branch and the shared pixel scale exhibits this particle zoo rather than some unrelated alternative one.

1.2 Logical basis and reading rule

The particle derivation uses the same OPH basis as the SM/GR derivation in Ref. [4]. Its five axioms are the screen net, overlap consistency, local MaxEnt with refinement stability, recoverable generalized entropy, and Minimal Admissible Realization (MAR). On that basis, the SM/GR derivation in Ref. [4] supplies the structural chain used here: compact gauge reconstruction, the realized Standard Model quotient, the hypercharge lattice, and the counting chain $N_g = 3$, $N_c = 3$. Refs. [2, 3] supply the patch-net, repair, measurement, and regulated screen language used later when flavor transport and observer-facing readout enter.

The quantitative particle side adds only two declared continuous inputs. The common pixel constant P feeds the forward D10 electroweak map and its descendants. The separate screen-capacity input N_{scr} belongs to the cosmological-capacity branch and only to legacy capacity-level neutrino side estimates. Structural carriers, calibration outputs, exact sidecars, and continuation lanes are therefore downstream branches of one common OPH premise ledger. Section 5 records that ledger in full.

2 Conceptual Overview: From S^2 Overlap Obstructions to Particle Families

Before the hard derivation chapters, it helps to say in plain but still technical language what the OPH particle derivation is trying to do. The basic picture is not "put particles on a sphere." The picture is: start with a network of observer patches on a screen S^2, ask how their local descriptions can fail to glue perfectly, and then study which persistent obstruction patterns survive repair, transport, and refinement. Those persistent patterns are what eventually harden into gauge sectors, flavor structure, and particle families.

2.1 Step 1: local observer patches do not automatically agree

Imagine a finite network of observer patches covering the screen. Each patch carries local data, and every neighboring pair of patches shares an overlap through which they can compare notes. If all local descriptions matched perfectly on every overlap, then there would be no nontrivial structure to study: the global state would be one big consistent codeword. But generic local descriptions do *not* line up so simply. They can disagree on overlaps, they can require local repair maps, and

they can carry nontrivial loop data when one transports information around cycles in the patch network.

This is where the patch-net language of Ref. [2], with inconsistency potential, local repair, cycle holonomy, and frustration becomes useful. Local agreement on one edge does not guarantee global agreement around a loop. So the first genuinely interesting objects in the OPH story are not yet particles. They are overlap defects, repair obstructions, and transport inconsistencies on the S^2 observer network.

2.2 Step 2: stable overlap obstructions become sector data

Once those overlap obstructions are organized algebraically, they stop looking like raw mismatches and start looking like sector data. The collar and edge-center structure tells us that boundary-visible information decomposes into sector labels, and the gauge-as-gluing perspective reinterprets part of what looked like mismatch bookkeeping as redundancy or implementation hiding. In other words, some of the nontrivial overlap structure is the mathematical content that later becomes gauge structure.

At this stage the key idea is: if sector labels can be transported coherently under refinement, and if their fusion and duality data satisfy the required categorical properties, then one can reconstruct a compact gauge group from them. This is the first big jump from screen-network obstructions to familiar particle-physics structure. The obstruction story has been compressed into transportable edge sectors, and those edge sectors become the raw material for gauge symmetry.

2.3 Step 3: the realized Standard Model branch fixes the structural carrier skeleton

Once compact gauge reconstruction is in place, MAR selects the realized Standard Model branch. That is where the particle-spectrum story begins to resemble ordinary particle physics. The realized branch fixes

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3,$$

and from there several carrier statements are structural. The photon, gluons, and graviton are not late numerical accidents. They are the symmetry-protected massless carriers of the realized electromagnetic, color, and dynamical-metric branches. So one direct route from screen-network obstruction language to particle content is:

$$\begin{aligned}&\text{overlap obstruction} \to \text{edge-sector transport} \to \text{compact gauge reconstruction}\\ &\qquad\qquad \to \text{realized Standard Model branch} \to \text{structural carrier content.}\end{aligned}$$

2.4 Step 4: refinement-stable transport on the three-generation bundle becomes flavor structure

The next jump is subtler and more interesting. The Standard Model branch tells us that three generations are realized, but it does not yet tell us what distinguishes one family from another. That is where the flavor derivation enters. Its starting point is still obstruction language, but in a more refined form: family transport kernels, generation-bundle branch generators, same-label eigenline transport, overlap-edge cocycles, projector data, spectral gaps, pair suppressions, and cycle phases.

Semi-informally, the idea is that persistent transport defects on the three-generation charged bundle organize themselves into a family observable. Once that object exists, it can be pushed

forward into common sector-response data for the up/down quark sectors, the charged-lepton sector, and the neutrino sector. In that sense, the particle families are not attached to the screen by hand. They are supposed to emerge from how the refinement-stable observer network stores and transports nontrivial overlap structure across the realized three-generation bundle.

2.5 Step 5: the same family backbone branches into different particle families

Once a common family object exists, the downstream sectors stop being arbitrary lists and become different readout branches of the same deeper transport structure. On the D12 surface studied here those branches are organized by one shared excitation dictionary rather than by independent lane-specific ansätze. The proof-facing same-label data are the gap/overlap scalars

$$(g_e, \omega_e), \qquad d_e := 1 - \omega_e, \qquad q_e := \sqrt{g_e d_e}, \qquad \eta_e := \log q_e - \frac{1}{3}\sum_f \log q_f,$$

equivalently the centered eta-class $[\eta_e]$ or the normalized weights

$$\mu_e := \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

Let s_e denote the realized sector-charge step carried by the same-label arrow e. For any realized readout path γ on the same-label transport graph, the flavor-side integers are the traversal multiplicities

$$m_{\gamma,e} \in \mathbb{Z}_{\geq 0}, \qquad s_\gamma = \sum_e m_{\gamma,e} s_e,$$

and the corresponding excitation action is

$$A_\gamma = -\sum_e m_{\gamma,e} \log q_e.$$

So the old one-number "texture exponents" are not primitive family data. They are only compressions of excitation action, for example

$$n_\gamma^{(6)} := A_\gamma / \log 6,$$

and they become literal defect counts only after the additional uniform sixfold collapse compatible with the realized $\mathbb{Z}_6$ quotient. The structural result fixed before those branches is still the realized counting chain $N_g = 3$ then $N_c = 3$. The current flavor boundary is not a fresh set of primitive flavor parameters, but the lane-specific readout closures above that dictionary.

Bosons. The bosonic branches split early. Structural massless carriers follow from the realized gauge and gravity branches. The W boson and Z boson then appear on the electroweak calibration stage, while the Higgs boson appears on the Higgs/top critical stage.

Quarks. Quarks are the first matter family with a selected-class exact public theorem. The quark derivation takes the shared flavor data, performs the mean split and sector descent, builds forward Yukawa surfaces, and closes the exact PDG 2025 API running-quark sextet together with explicit exact forward Yukawas on the selected public quark frame class f_P. The mass-side route compresses transported same-label data to $(\Delta_{ud}^{\text{overlap}}, \eta_Q^{\text{centered}})$ and, on the compact same-family branch, to the emitted ray $\mathcal{R}_{D12}^{ud}$. The top quark also remains visible on the Higgs/top critical stage. On the

mixing side the historical same-label left-handed orbit closes negatively at the singleton σ_{ref}, but the public theorem is stronger: the selected public quark frame class chosen by P carries the exact physical sigma datum, the affine mean law emits (g_u, g_d) algebraically, the ordered three-point readout emits the exact running sextet, and the exact forward construction emits explicit exact forward Yukawas Y_u and Y_d. The quark lane therefore splits into four linked surfaces: the exact same-family witness on the selected D12 sheet, a restricted common-refinement transport-frame exact chain that carries the same sextet together with explicit forward Yukawas Y_u and Y_d on its declared carrier, the target-free mass bridge `quark_d12_t1_value_law` on the emitted D12 mass ray, and the selected-class public theorem on f_P. This is selected-class closure only. It does not claim a global classification of all quark frame classes.

Charged leptons. Charged leptons are not described here as a Koide-assisted fit. The exact statement is narrower. On the ordered charged carrier

$$(-1, x_2, 1), \qquad x_2 = -0.5175863354681689,$$

every charged source pair (η, σ) determines the centered charged logs by

$$e_{\mathrm{log,centered}} = -\frac{(3+x_2)\sigma - \eta}{6}, \qquad \mu_{\mathrm{log,centered}} = \frac{x_2\sigma - \eta}{3}, \qquad \tau_{\mathrm{log,centered}} = \frac{(3-x_2)\sigma + \eta}{6}.$$

So the centered charged hierarchy closes once a charged source pair is emitted, but the public mass rows do not.

The remaining theorem-grade frontier is explicitly two-layered:

$$\widehat{C}_e^{\mathrm{cand}} \longrightarrow \widehat{C}_e \quad \Longrightarrow \quad \mu_{\mathrm{phys}}(Y_e) \quad \Longrightarrow \quad (g_e, \Delta_e^{\mathrm{abs}}).$$

The first arrow is the package `oph_generation_bundle_branch_generator_splitting`. It is blocked by the missing descended-commutator clause `compression_descendant_commutator_vanishes_or_is_uniformly_quadratic_small_after_central_split`, because the corpus proves neither exact vanishing nor uniform quadratic smallness after the central split. Even a successful closure at this first layer would promote only the centered operator candidate. The second arrow concerns the common-shift quotient surface, which is closed negatively, so centered data alone do not emit an absolute charged scale. The post-promotion slot is the refinement-stable uncentered trace lift, and the exact smaller forcing object beneath $\mu_{\mathrm{phys}}(Y_e)$ is `charged_physical_identity_mode_equalizer`, namely the physical identity-mode equalizer $\delta(r, r') = 0$ on common physical fibers. From that scalar the uncentered lift, determinant-line section, and affine anchor follow canonically,

$$\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\mathrm{phys}}(Y_e)\,\mathbf{1}, \qquad s_{\det}(Y_e) = 3\mu_{\mathrm{phys}}(Y_e), \qquad A_{\mathrm{ch}}(Y_e) = \mu_{\mathrm{phys}}(Y_e).$$

This second layer is a descent contract rather than a second fitted invariant: if theorem-grade $\widehat{C}_e$ and an admissible refinement-stable uncentered lift exist, the physical identity-mode equalizer forces the descended scalar $\mu_{\mathrm{phys}}(Y_e)$. The public claim is exact same-carrier centered readback together with a closed common-shift no-go and the explicit two-layer remaining obstruction above. Any balanced-circulant / $Q = 2/3$ / $\delta = 2/9$ branch is compare-only continuation and not an emitted charged-lepton mass theorem.

Neutrinos. The isotropic intrinsic neutrino branch is excluded by the exact atmospheric-splitting cap, so the live continuation lane is the weighted-cycle / Majorana-holonomy branch. On that lane the standard-math-fixed balanced transport-load selector is the arithmetic midpoint on the positive segment between $\chi = 1 + \varepsilon$ and $1 + \gamma/2$,

$$D_\nu = \frac{\chi + 1 + \gamma/2}{2}, \qquad p_\nu(\gamma, \varepsilon) = 1 + \gamma + \frac{\varepsilon}{D_\nu}.$$

This repair closes the oscillation-shape lane in the physical PMNS/hierarchy regime. The emitted weighted-cycle data are

$$\theta_{12} = 34.225904631810025^\circ, \qquad \theta_{23} = 49.72282845058266^\circ, \qquad \theta_{13} = 8.686355527700156^\circ,$$

$$\delta_{\mathrm{PMNS}} = 305.58061231449796^\circ, \qquad J = -0.02753115613565372, \qquad \frac{\Delta m_{21}^2}{\Delta m_{32}^2} = 0.03072111.$$

So no further discrete selector remains on the neutrino lane.

The normalized same-label overlap-defect weight section closes beneath that shape. The paper-facing amplitude parameterization is

$$B_\nu := \lambda_\nu q_{\mathrm{mean}}^{p_\nu} / m_{\star,\mathrm{eV}}$$

above the closed normalizer, while the emitted proxy is

$$P_\nu := I_\nu^{1/2} \widehat{\mathrm{ratio}}^{\,1/2} \,\mathrm{sum_defect}^{-1} = 6.699825740519345.$$

Writing

$$G := \mathrm{sum_gap}, \qquad Q := \mathrm{prod_qbar}, \qquad S := \mathrm{solar_response_over_mstar},$$

the weighted-cycle bridge rigidity theorem emits

$$C_\nu = G^2 Q S^{-1/2} = \mathrm{sum_gap}^2\, \mathrm{prod_qbar}\, \mathrm{solar_response_over_mstar}^{-1/2} = 0.9994295999075177.$$

Therefore

$$B_\nu = P_\nu\, C_\nu = 6.696004159297337, \qquad \lambda_\nu = \frac{m_{\star,\mathrm{eV}}}{q_{\mathrm{mean}}^{p_\nu}}\, P_\nu\, C_\nu = 1.7237014208357415.$$

If $\hat{m}_i$ and $\widehat{\Delta m_{ij}^2}$ denote the branch-fixed scale-free eigenmasses and splittings, then

$$m_i = \lambda_\nu\, \hat{m}_i, \qquad \Delta m_{ij}^2 = \lambda_\nu^2\, \widehat{\Delta m_{ij}^2},$$

so every PMNS observable and every dimensionless hierarchy ratio propagate to one emitted absolute family. The best closed constructive local object beneath that reduced scalar is the defect-weighted same-label edge family

$$q_e = \sqrt{g_e d_e}, \qquad \eta_e = \log q_e - \frac{1}{3} \sum_f \log q_f, \qquad \mu_e = \mu_\nu\, \frac{e^{\eta_e}}{\frac{1}{3} \sum_f e^{\eta_f}},$$

but that family does not emit C_ν or the induced B_ν by itself. The emitted absolute family is

$$(m_1, m_2, m_3) = (0.017454720257976796,\ 0.019481987935919015,\ 0.05307522145074924)\,\mathrm{eV},$$

with

$$\Delta m_{21}^2 = 7.488059465106851 \times 10^{-5}\,\mathrm{eV}^2,$$
$$\Delta m_{31}^2 = 2.5123118727618473 \times 10^{-3}\,\mathrm{eV}^2,$$
$$\Delta m_{32}^2 = 2.4374312781107786 \times 10^{-3}\,\mathrm{eV}^2.$$

The exact two-parameter positive-segment adapter, the bridge corridor, and the reduced-correction candidate audit remain on disk only as diagnostic surfaces and do not feed back into theorem state. Charged leptons and neutrinos are split for different reasons. Charged leptons carry exact same-carrier readback and a closed common-shift no-go, then wait on promotion of $\widehat{C}_e^{\mathrm{cand}}$ and on the post-promotion lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$, with the physical identity-mode equalizer as the exact smaller forcing object beneath that scalar; the weighted-cycle neutrino branch carries the emitted bridge-rigid absolute family. Taken together with the exact W/Z frozen repair pair, the closed Higgs/top forward seed together with its compare-only exact inverse sidecar, the exact charged/quark same-family witnesses, and the emitted weighted-cycle neutrino bridge-rigid absolute family, the exact non-hadron output lane is complete, on a mixed theorem/compare-only/same-family surface rather than one uniform theorem tier.

Claim-tier boundary. The Stage-5 summary is this: quarks close on the selected public quark frame class f_P; charged leptons have an exact same-carrier readback theorem together with an exact absolute-scale obstruction; neutrinos carry an emitted weighted-cycle bridge-rigid absolute family above the scale-free branch. None of those later-family statements belongs on the recovered-core surface fixed earlier by the gauge and relativity chain.

2.6 Structural results and open closures

The move from overlap structure to compact gauge reconstruction and the realized Standard Model branch belongs to the structural core. The move from the realized branch to structural carriers and to the electroweak/Higgs-top bosonic outputs is reflected in the reported numerical rows. The move from the shared flavor dictionary to full family closure is split by sector.

Quarks close on the selected public quark frame class f_P. The lower theorem-emitted quark package on the declared ledger consists of the D12 mass ray $\mathcal{R}_{D12}^{ud}$, the negative same-label left-handed selector closure $\sigma_{ud} = \sigma_{\mathrm{ref}}$, and the restricted-scope affine mean package with $g_{\mathrm{ch}} = 0.9231656602589082$ on `shared_budget_only` and $(g_u, g_d) = (0.7797392875757557, 0.12172551081512113)$ on `current_family_only`. Separate target-free theorems internalize the mass bridge $\Delta_{ud}^{\mathrm{overlap}} = \frac{1}{6}\log(c_d/c_u)$, equivalently $\Theta_{ud}^{\mathrm{mass}} =$ `quark_d12_t1_value_law`, on the emitted D12 ray. The selected-class public theorem is the direct descent statement on f_P: the exact physical sigma datum is target-free public on that selected class, the affine mean law emits (g_u, g_d) algebraically, the ordered three-point quadratic readout emits the exact sextet, and the exact forward construction emits explicit exact forward Yukawas Y_u and Y_d. A separate D12 internal backread sidecar also fixes the mass-side scalar package numerically on a continuation-only surface, with $\Delta_{ud}^{\mathrm{overlap}} = 0.13431740757663183$, $t_1 = 0.6715870378831591$, $\eta_Q^{\mathrm{centered}} = -0.018210067243314802$, and $\kappa_Q = -0.012436796997836279$; that sidecar does not replace the selected-class public theorem and does not classify all quark frame classes. Charged leptons sit on a narrower continuation surface: the same-carrier centered readback is exact, the common-shift quotient surface is closed as a no-go for absolute scale, and the theorem-grade frontier is the two-layer chain

$$\widehat{C}_e^{\mathrm{cand}} \longrightarrow \widehat{C}_e \quad\Longrightarrow\quad \mu_{\mathrm{phys}}(Y_e) \quad\Longrightarrow\quad (g_e, \Delta_e^{\mathrm{abs}}),$$

with the first arrow identified with the package `oph_generation_bundle_branch_generator_splitting` and reduced in practice to the missing descended-commutator clause `compression_descendant_commutator_vanishes_or_is_uniformly_quadratic_small_after_central_split`. The post-promotion slot is the refinement-stable uncentered trace lift, and within that lift the exact smaller forcing object beneath $\mu_{\mathrm{phys}}(Y_e)$ is `charged_physical_identity_mode_equalizer`, equivalently the physical identity-mode equalizer $\delta(r, r') = 0$ on common physical fibers. If that lift closes, the uncentered lift, determinant-line section, and affine anchor follow canonically from $\mu_{\mathrm{phys}}(Y_e)$; until then no public charged-lepton mass theorem is claimed. A target-anchored same-family exact witness reproduces the charged reference triple exactly on the same ordered eigenvalue family, and within that same-family scope the ordered three-point quadratic readout is closed through `oph_lepton_current_family_quadratic_readout_theorem`. On that same fixed target-anchored witness, the scoped affine coordinate also closes through `oph_lepton_current_family_affine_anchor_theorem`, with $A_{\mathrm{ch}}^{(\mathrm{sf})} = \frac{1}{3}\log\det(Y_e^{(\mathrm{sf})})$ and $g_e^{(\mathrm{sf})} = e^{A_{\mathrm{ch}}^{(\mathrm{sf})}}$; this remains same-family-only and does not promote the theorem lane. Neutrinos close on a different theorem surface: the weighted-cycle / Majorana-holonomy lane fixes the PMNS/hierarchy shape, and the bridge-rigidity theorem emits

$$C_\nu = G^2 Q S^{-1/2} = 0.9994295999075177,$$

with $G = \mathrm{sum_gap}$, $Q = \mathrm{prod_qbar}$, and $S = \mathrm{solar_response_over_mstar}$. The emitted proxy is

$$P_\nu = 6.699825740519345,$$

the induced paper-facing amplitude parameterization is

$$B_\nu = P_\nu C_\nu = 6.696004159297337,$$

and the absolute-attachment theorem then fixes

$$\lambda_\nu = \frac{m_{\star,\mathrm{eV}}}{q_{\mathrm{mean}}^{p_\nu}} P_\nu C_\nu = 1.7237014208357415.$$

hence the absolute family

$$(m_1, m_2, m_3) = (0.017454720257976796,\ 0.019481987935919015,\ 0.05307522145074924)\ \mathrm{eV},$$

$$\begin{aligned}
\Delta m_{21}^2 &= 7.488059465106851 \times 10^{-5}\ \mathrm{eV}^2,\\
\Delta m_{31}^2 &= 2.5123118727618473 \times 10^{-3}\ \mathrm{eV}^2,\\
\Delta m_{32}^2 &= 2.4374312781107786 \times 10^{-3}\ \mathrm{eV}^2.
\end{aligned}$$

The exact two-parameter positive-segment adapter, the bridge corridor, and the reduced-correction candidate audit remain on disk only as diagnostic sidecars beneath the bridge-rigidity invariant C_ν and its induced paper-facing amplitude B_ν. Hadrons remain behind the explicit nonperturbative frontier.

The conceptual picture stays the same across the later sector chapters. Particle families are the stabilized and readable descendants of transport obstructions on an observer network over S^2, and each chapter records how far that picture has been turned into explicit mathematics and reported outputs. None of those family-continuation statements belongs to the recovered-core surface fixed earlier by the gauge and relativity chain.

3 Calibration and the Reverse-Engineering Strategy

Before the claim-tier table, it is worth stating what the reverse-engineering claim actually is. In ordinary phenomenological use, the Standard Model does not derive the observed particle masses and mixings from one common microscopic constant. It is usually presented with 19 free parameters in the minimal massless-neutrino theory, and with at least 26 once neutrino masses and mixing are included, depending on neutrino-sector conventions. The OPH particle derivation is trying to compress that situation radically. The aim is not to feed in one Yukawa coupling, one mass, or one mixing angle per sector, but to infer one universal pixel constant P from a small calibration family and then demand that the same P drive all downstream bosonic, quark, lepton, neutrino, and hadronic branches.

One clarification matters. The broader formulation carries the screen-capacity input N_{scr} in its cosmological branch. So the claim here is not that every continuous number appearing anywhere in the broader paper stack has disappeared. The claim is narrower and more interesting: the particle-spectrum reverse-engineering derivation is organized around one common nontrivial scale input P, instead of a long particle-by-particle parameter list.

3.1 Why calibration is necessary

Semi-informally, the structural OPH theorems tell us *what kind* of particle world is realized, but not yet *what absolute scale in GeV* that world occupies. The overlap, modular, gauge, anomaly, and admissibility arguments recover the realized Standard Model branch, the exact hypercharge pattern, three generations, three colors, and the symmetry-protected massless carriers. But those structural results do not by themselves tell us the numerical values of the W boson mass, the Higgs boson mass, or the up-quark mass. Calibration is the bridge from the screen-side structural world to observed laboratory units.

Mathematically, the calibration problem can be phrased as follows. Let

$$\mathcal{C} = \{O_i^{\mathrm{obs}}\}$$

be a small family of observed calibration quantities, and let

$$O_i^{\mathrm{pred}} = F_i(P)$$

be the forward OPH prediction for each such quantity when the pixel constant is set to P. Then each calibration observable defines an implied pixel constant

$$P_i := F_i^{-1}(O_i^{\mathrm{obs}}),$$

whenever the inverse is locally meaningful. Exact single-P closure means that the same value

$$P_* \quad \text{satisfies} \quad P_i = P_* \text{ for all calibration observables.}$$

Once such a common P_* exists, the real predictive question becomes whether all downstream particle readouts can be written as

$$X_j = G_j(P_*)$$

with no new sector-specific fit constants inserted by hand.

3.2 How the D10 electroweak branch works

In the implementation used here, the quantitative D10 branch is electroweak. Its public target-free output family is

$$\{m_W,\ m_Z,\ \alpha_{\rm em}(m_Z),\ \sin^2\theta_W(m_Z),\ v\},$$

emitted from one declared input P on the printed running/matching package. The construction proceeds as follows: start from P, build the single-P running electroweak family, reduce it to the two-scalar carrier $(\sigma_{\rm EW}, \eta_{\rm EW})$, apply the carrier selector on that reduced chart, and read off the W boson/Z boson mass pair from the selected carrier point. The Higgs/top critical stage then inherits this D10 electroweak core and does not introduce a new free-input sector. The forward logical order on the D10 surface is:

$$P \longmapsto \big(M_U(P), E_{\rm cell}(P)\big) \longmapsto \alpha_U(P) \longmapsto \big(t_U(P), t_{\rm tr}(P)\big) \longmapsto \big(t_2(P), t_3(P), v(P)\big) \longmapsto \alpha_i(\mu_*; P).$$

The forward transmutation certificate records that the same source-only basis reconstructs the unified diffusion parameter $t_U(P) = 4\pi^2\alpha_U(P)$ and transmutation exponent $t_{\rm tr}(P) = 2\pi/((N_c + 1)\alpha_U(P))$ as the pixel-closure solve itself. Measured electroweak data are not used to infer those internal transmutation parameters; they appear only at the end as compare-only validation targets.

This is why the D10 branch is not a side detail. It is the place where the particle derivation shows whether one common input P really emits the electroweak rows or merely hides many fitted inputs behind fancy notation. In the electroweak branch, the public W/Z rows are emitted by a target-free source-only repair theorem from the source basis fixed by the forward D10 solve. The derivation also contains two auxiliary validation surfaces: the exact current-carrier chart and the freeze-once coherent repair surface. On the calibration tier, the electroweak mass-side lane has no open blocker.

Throughout this paper, labels such as $D10$, $D12$, $D12^{\rm mass}_{ud}$, and $\mathcal{R}^{ud}_{D12}$ are internal OPH ledger labels inherited from [4], not external references. In that ledger, $D10$ names the Phase-II electroweak calibration branch and $D12$ names the Phase-III phenomenological-continuation branch. The public paper ledger has no separate D11 row. The overview/dependency table in Ref. [4] is the canonical key, and this paper uses those labels only to locate each particle-side object inside the broader OPH dependency map.

3.3 What the D10 electroweak branch fixes

On the declared electroweak running/matching surface, the external pixel input P fixes the source basis $(\alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\rm source}, v)$. The target-free source-only repair theorem emits one coherent quintet $(W, Z, \alpha_{\rm em}^{-1}, \sin^2\theta_W, v)$ from that source basis fixed by the forward solve. The derivation also includes two explicit validation surfaces beneath the public theorem output: the exact current-carrier chart and the freeze-once coherent repair surface. The same tiered logic is useful later in the paper because flavor and neutrino derivations do have open theorem objects, whereas the electroweak calibration lane does not. On the sharpened D10 ledger, the paper-side transmutation factor is $\beta_{\rm EW} = N_c + 1$; overloaded β-ratios appear only on compare-only diagnostic readouts and are not part of the theorem contract.

4 Results at a Glance

In plain language, the particle derivation has three different kinds of rows: rows that are structurally fixed, rows that are numerically emitted, and rows whose chains stop short of a promotable mass prediction. This section summarizes that split.

Read this section by claim tier. The same absolute residual can indicate either a theorem-grade emitted value, an exact sidecar, or a compare-only continuation benchmark. On this table, the structural zeros are exact on their declared surfaces; the electroweak lane carries forward-emitted public rows plus an exact frozen sidecar; the Higgs, charged, and quark lanes each carry a stronger exact sidecar than the theorem lane beneath them; and the weighted-cycle neutrino lane carries the bridge-rigidity invariant C_ν above the emitted proxy P_ν, with B_ν retained only as the induced paper-facing amplitude, while the older exact adapter remains diagnostic-only.

Sector	Claim tier	Derivation stage	Public output(s) / exact sidecar(s)	Caveat / remaining step
Structural carriers	structural	gauge/gravity structural theorems	exact $m_\gamma = m_g = m_{\mathrm{grav}} = 0$	theorem-grade structural exactness; no particle-side caveat
Electroweak bosons	calibration	target-free D10 electroweak branch	public target-free W/Z rows together with the exact frozen-repair pair $M_W = 80.377$ GeV, $M_Z = 91.18797809193725$ GeV	exact only on the frozen authoritative repair surface; the public theorem rows remain the target-free source-only D10 outputs
Higgs/top stage	secondary quantitative + closed forward seed + exact sidecars	D10 gauge core followed by D11 forward seed / Jacobian / family exact witnesses	public forward pair $m_H = 125.218922$ GeV, $m_t = 172.388646$ GeV; exact inverse-sidecar pair on the D11 Jacobian	the public D11 forward seed is closed on its emitted one-scalar branch; the inverse pair is compare-only and does not replace the forward rows
Quark family	selected-class theorem + exact supporting surfaces	shared excitation dictionary → D12 mass bridge → selected public quark frame class f_P → direct public sigma-datum descent → affine mean law → ordered three-point readout → exact forward construction	exact running quark sextet on the selected public quark frame class f_P, together with explicit exact forward Yukawas Y_u, Y_d on that selected class	selected-class closure only. The exact sextet matches the official PDG 2025 API running-quark surface, with the top coordinate taken from the PDG cross-section entry. Supporting exact surfaces: same-family witness on `current_family_only`; restricted common-refinement transport-frame theorem on `current_family_common_refinement_transport_frame_only`. Both realize the same sextet and explicit exact forward Yukawas. Separate target-free mass bridge: $\Delta_{ud}^{\mathrm{overlap}} = \frac{1}{6}\log(c_d/c_u)$, equivalently $\Theta_{ud}^{\mathrm{mass}} =$ `quark_d12_t1_value_law`, on the emitted D12 mass ray. Historical lower selector: σ_{ref} is a negative same-label sheet selector statement beneath the selected-class theorem. No global classification of all quark frame classes is claimed
Charged leptons	continuation + exact sidecar witness	shared excitation dictionary → ordered charged carrier → exact centered readback	exact same-carrier centered readback together with the exact same-family charged triple on a closed ordered-three-point readout chain	same-family-only on the exact witness. The centered common-shift quotient is closed negatively. Theorem-grade waiting set is promotion of $\widehat{C}_e^{\mathrm{cand}}$. Then the post-promotion lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$, with the physical identity-mode equalizer beneath that scalar
Neutrinos	continuation + emitted bridge invariant + diagnostic sidecars	same-label scalar certificate → weighted-cycle branch → bridge rigidity → absolute attachment	emitted absolute masses and central splittings on the weighted-cycle branch, with bridge invariant C_ν above emitted proxy P_ν and paper-facing amplitude B_ν	theorem-grade on the declared weighted-cycle branch; the exact positive-segment adapter, bridge corridor, and correction audit remain diagnostic-only and do not feed back into theorem state
Hadrons	execution-contract-frozen nonperturbative continuation	stable-channel correlator and readout derivation	no paper-derived hadron masses	outside the declared particle roadmap: one production backend export bundle with publication-complete manifest provenance, real correlator arrays, and production systematics, followed by large-scale nonperturbative execution that is not a realistic target within the current project compute scope

Two points are worth stating explicitly. First, the theorem rows and the exact-hit rows are not the same object. The theorem table carries selected-class quark closure and an open charged boundary, while the exact-hit surface above it also contains exact same-family charged and quark witnesses, the restricted transport-frame quark exact chain, and exact diagnostic sidecars. Second, "the bosons are finished" is too blunt. The photon, gluons, and graviton are structural zeros; the W boson and Z boson come from electroweak calibration; and the Higgs boson sits on the Higgs/top

critical stage. Those are all real outputs, but they live at different tiers.

This claim-tier boundary governs the rest of the paper. Whenever a later chapter becomes detailed about an open family, it should be read through this table. The table records the constructive chain and keeps its claim tier explicit.

Exact non-hadron hit surface. If one asks only for exact hits, the paper-facing non-hadron bundle splits by lane as follows:

Lane	Exact output(s)	Exact OPH chain on the paper surface	Caveat
Structural carriers	$m_\gamma = m_g = m_{\mathrm{grav}} = 0$	axioms → realized electromagnetic/color/dynamical-metric carrier skeleton → symmetry-protected zeros	theorem-grade structural exactness
Electroweak exact sidecar	exact frozen W/Z pair	axioms → $D7$ → $D10$ calibration chain → exact frozen repair pair `oph_d10_ew_w_anchor_neutral_shear_factorization`	exact only on the frozen authoritative repair surface; compare-only beneath the target-free theorem
Higgs/top exact sidecar	exact Higgs/top inverse pair	axioms → $D10$ gauge core → $D11$ Jacobian → exact inverse slice	compare-only inverse slice; the public D11 rows come from the closed one-scalar forward seed
Charged exact witness	exact same-family charged triple	axioms → shared excitation dictionary → ordered charged carrier → exact centered readback → closed quadratic readout theorem → same-family exact witness	same-family-only; theorem lane carries exact centered readback plus the closed common-shift no-go, then waits on $\widehat{C}_e^{\mathrm{cand}}$ and the post-promotion lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$, with the physical identity-mode equalizer beneath that scalar
Quark selected-class theorem	exact running quark sextet plus explicit exact forward Yukawas on f_P	axioms → shared excitation dictionary → D12 mass bridge → selected public quark frame class f_P → direct public sigma-datum descent → affine mean law → ordered three-point readout → exact forward construction	selected-class closure on the public physical quark frame class chosen by P; the sextet matches the official PDG 2025 API running-quark surface exactly, with the top coordinate taken from the PDG cross-section entry rather than the auxiliary direct-top entry; the same exact sextet is also realized by the same-family witness on `current_family_only` and by the restricted common-refinement transport-frame exact chain, which also carries explicit exact forward Yukawas Y_u and Y_d; the target-free mass bridge $\Delta_{ud}^{\mathrm{overlap}} = \frac{1}{6}\log(c_d/c_u)$, equivalently $\Theta_{ud}^{\mathrm{mass}} =$ `quark_d12_t1_value_law`, closes separately on the emitted D12 mass ray; the historical selector σ_{ref} is a lower negative sheet statement beneath the selected-class theorem; no global classification of all quark frame classes is claimed
Neutrino theorem branch	emitted absolute mass triple and representative central splittings	axioms → same-label scalar certificate → weighted-cycle branch → bridge-rigidity theorem → absolute-attachment theorem	theorem-grade on the declared weighted-cycle branch; the exact positive-segment adapter and bridge-coordinate sidecars remain diagnostic-only

Concretely, the exact values shown on this surface are

$$(M_W, M_Z, m_H) = (80.377,\ 91.18797809193725,\ 125.1995304097179)\,\mathrm{GeV},$$

$$(m_e, m_\mu, m_\tau) = (0.00051099895,\ 0.1056583755,\ 1.7769324651340912)\,\mathrm{GeV},$$

$$(u, d, s, c, b, t) = (0.00216,\ 0.00470,\ 0.0935,\ 1.273,\ 4.183,\ 172.3523553288311)\,\mathrm{GeV},$$

which matches the official PDG 2025 API running-quark target surface exactly, with the top coordinate taken from the PDG cross-section mass entry rather than the auxiliary direct-top entry. On the declared current-family common-refinement transport-frame carrier, the restricted exact chain goes further and emits explicit exact forward Yukawas Y_u and Y_d. and

$$(m_1, m_2, m_3) = (0.017454720257976796,\ 0.019481987935919015,\ 0.05307522145074924)\,\mathrm{eV},$$

with emitted weighted-cycle absolute splittings

$$\begin{aligned}
\Delta m_{21}^2 &= 7.488059465106851 \times 10^{-5}\,\mathrm{eV}^2,\\
\Delta m_{31}^2 &= 2.5123118727618473 \times 10^{-3}\,\mathrm{eV}^2,\\
\Delta m_{32}^2 &= 2.4374312781107786 \times 10^{-3}\,\mathrm{eV}^2.
\end{aligned}$$

The table is lane-based. It states the OPH chain that emits each exact-hit surface and the exact caveat that prevents silent promotion to a stronger theorem claim.

Local unification surface. The bosonic rows carry one cross-lane statement. The same calibrated input P fixes the D10/D11 bosonic trunk

$$P \longmapsto \alpha_U(P) \longmapsto \big(t_U(P), t_{\mathrm{tr}}(P)\big) \longmapsto v(P) \longmapsto (M_W, M_Z)\,.$$

$$\sigma_{D11,\mathrm{HT}} = \alpha_U(P)\cos(2\theta_{W0})/\sqrt{\pi} \longmapsto (m_H, m_t)\,.$$

Here $\alpha_U(P)$ is the branch value selected by the same forward D10 pixel-closure solve, so the bosonic trunk contains no inverse electroweak readback of the internal transmutation data. On the companion gravity side, the same pixel law packages

$$\bar{\ell}_{\mathrm{SU}(2)}(t_{2,\mathrm{run}}) \ +\ \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\mathrm{run}}) = P/4, \qquad G = \frac{a_{\mathrm{cell}}}{4\,\bar{\ell}(t)}.$$

The local unification surface has five explicit boundary objects: exact public W/Z requires one target-free D10 chart-identity theorem, exact public Higgs requires one live-forward D11 promotion theorem, and the gravity-side release requires one shared edge-entropy bridge together with the local readout package and the strict classical-regime clause. On that declared extension surface the local release bundle is

$$c = 299792458\,\mathrm{m/s}, \qquad G = 6.674299995910528 \times 10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2},$$

$$(M_W, M_Z, M_H) = (80.377,\ 91.18797809193725,\ 125.1995304097179)\,\mathrm{GeV},$$

with c as the structural Lorentz output and with the G row as an exact emitted branch value rather than a literal zero-difference identity against the rounded benchmark 6.6743×10^{-11}.

The detailed inventory below records the public theorem/continuation rows sector by sector. Whenever an exact sidecar or same-family witness is stronger than the public theorem row, the caveat is the one stated in the exact-hit table above.

4.1 Detailed particle inventory

Family	Particle	Claim tier	OPH value or strongest derived benchmark	Remaining step
Structural carriers	photon	structural	0	none
Structural carriers	gluons	structural	0 for the color gauge sector	none
Structural carriers	graviton	structural	0 for the dynamical-metric spin-2 carrier	none
Electroweak D10 branch	W boson	calibration	80.37700001539531 GeV from the closed target-free D10 repair theorem on the source basis fixed by the forward solve	closed D10 electroweak row; freeze-once validation sidecar yields the exact W/Z pair
Electroweak D10 branch	Z boson	calibration	91.18797807794321 GeV from the same closed target-free D10 repair theorem	same closed D10 electroweak theorem; exact W/Z sidecar sits on the frozen validation surface
Higgs/top critical stage	Higgs boson	secondary quantitative	125.218922 GeV	present a proof package for the closed one-scalar forward seed
Quark family	top quark	secondary quantitative	172.388646 GeV from the Higgs/top critical stage	same one-scalar seed proof package as the Higgs row
Quark family	up quark	selected-class theorem	0.00216 GeV on the selected public quark frame class f_P	exact running-mass theorem row on f_P; supporting exact same-family and restricted transport-frame surfaces realize the same value; no global classification of all quark frame classes is claimed
Quark family	down quark	selected-class theorem	0.00470 GeV on the selected public quark frame class f_P	same selected-class theorem surface; target-free mass bridge on the emitted D12 ray closes separately
Quark family	strange quark	selected-class theorem	0.0935 GeV on the selected public quark frame class f_P	same selected-class theorem surface; the historical selector σ_{ref} is a lower negative sheet statement beneath that theorem
Quark family	charm quark	selected-class theorem	1.273 GeV on the selected public quark frame class f_P	same selected-class theorem surface; explicit exact forward Yukawas Y_u, Y_d are emitted on f_P
Quark family	bottom quark	selected-class theorem	4.183 GeV on the selected public quark frame class f_P	same selected-class theorem surface; the exact sextet matches the official PDG 2025 API running-quark target surface
Charged leptons	electron	continuation gap	n/a; exact same-carrier centered readback exists once a charged source pair is emitted, but the absolute scale is blocked by the closed common-shift no-go; compare-only target $g_e^\star = 0.0457789$, equivalently $\Delta_e^{\mathrm{abs},\star} = 3.00398633$	promote the latent candidate $\widehat{C}_e^{\mathrm{cand}}$ by closing branch-generator splitting, then emit the post-promotion lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$; within that lift the exact smaller forcing object is the physical identity-mode equalizer, after which $\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\mathrm{phys}}(Y_e)\,\mathbf{1}$, $A_{\mathrm{ch}}(Y_e) = \mu_{\mathrm{phys}}(Y_e)$, and the readouts g_e, Δ_e^{abs} follow
Charged leptons	muon	continuation gap	n/a; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\widehat{C}_e^{\mathrm{cand}} \to$ promotion $\to$ post-promotion lift $\to \mu_{\mathrm{phys}}(Y_e) \to A_{\mathrm{ch}} \to g_e$ closure chain, with the physical identity-mode equalizer beneath the descended scalar
Charged leptons	tau lepton	continuation gap	n/a; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\widehat{C}_e^{\mathrm{cand}} \to$ promotion $\to \mu_{\mathrm{phys}}(Y_e) \to A_{\mathrm{ch}} \to g_e$ closure chain, with the physical identity-mode equalizer beneath the descended scalar
Neutrinos	ν_e	weighted-cycle theorem branch	0.017454720257976796 eV; the weighted-cycle branch emits $\theta_{12} = 34.2259^\circ$, $\theta_{23} = 49.7228^\circ$, $\theta_{13} = 8.68636^\circ$, $\delta = 305.581^\circ$, $J = -0.02753$, $\Delta m_{21}^2/\Delta m_{32}^2 = 0.03072111$, the bridge invariant $C_\nu = 0.9994295999075177$, the emitted proxy $P_\nu = 6.699825740519345$, the paper-facing amplitude $B_\nu = 6.696004159297337$, and $\lambda_\nu = 1.7237014208357415$	closed on the declared weighted-cycle theorem branch; the positive-segment adapter, bridge corridor, and bridge-coordinate sidecars remain diagnostic-only and do not feed back into theorem state
Neutrinos	ν_μ	weighted-cycle theorem branch	0.019481987935919015 eV; same weighted-cycle theorem branch as the electron-neutrino row	same bridge-invariant / induced-amplitude package and same diagnostic-only retirement rule for the positive-segment adapter and bridge corridor
Neutrinos	ν_τ	weighted-cycle theorem branch	0.05307522145074924 eV; same weighted-cycle theorem branch as the electron-neutrino row	same bridge-invariant / induced-amplitude package and same diagnostic-only retirement rule for the positive-segment adapter and bridge corridor
Hadrons	proton	execution-contract-frozen	not a paper-derived proton row; surrogate finest-ensemble N_{iso} candidate 0.938960210578 GeV and production geometry summary with 3 ensembles / 6 cfg	outside the declared particle roadmap: replace the surrogate bridge with one production backend export bundle plus production systematics, requiring large-scale nonperturbative execution beyond the current project compute scope

Family	Particle	Claim tier	OPH value or strongest derived benchmark	Remaining step
Hadrons	neutron	execution-contract-frozen	not a paper-derived neutron row	outside the declared particle roadmap: production backend export bundle plus isospin-resolved hadron systematics, requiring large-scale nonperturbative execution beyond the current project compute scope
Hadrons	neutral pion proxy	execution-contract-frozen	not a paper-derived pion row; surrogate finest-ensemble π_{iso} candidate 0.135039383836 GeV	outside the declared particle roadmap: replace the surrogate bridge with one production backend export bundle plus production systematics, requiring large-scale nonperturbative execution beyond the current project compute scope
Hadrons	$\rho(770)^0$ proxy	execution-contract-frozen	not a paper-derived resonance row	outside the declared particle roadmap: production backend bundle plus finite-volume resonance extraction, requiring large-scale nonperturbative execution beyond the current project compute scope

5 Detailed Premise Ledger, Screen Architecture, Inputs, and Theorem Packages

This section records the detailed premise ledger used by the particle derivation: the five axioms, the two declared external continuous inputs, the regulated screen realization, and the imported OPH theorem packages. The order is the logical order used later in the paper: axioms and screen language first, external inputs next, and theorem packages after that.

5.1 Canonical OPH basis

The particle-spectrum derivation uses the same five axioms as the compact reconstruction paper [4]. They are restated here for local readability because every later particle-family chapter depends on them either directly or through the structural OPH chain imported from that paper.

Axiom 5.1 (Screen Net)**.** *Physical data is organized on a horizon screen S^2 carrying a net of local algebras*

$$P \longmapsto \mathcal{A}(P)$$

for connected patches $P \subset S^2$, with isotony

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

Axiom 5.2 (Overlap Consistency)**.** *For overlapping patches $P_1 \cap P_2 \neq \varnothing$, the local states induced on the shared algebra agree:*

$$\omega_{P_1}|_{\mathcal{A}(P_1 \cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1 \cap P_2)}.$$

Axiom 5.3 (Local MaxEnt and Refinement Stability)**.** *At the regulator scale ℓ_{UV}, the realized branch is selected by maximizing entropy subject to a finite family of gauge-invariant local constraints*

$$\mathcal{C}_{\ell_{\mathrm{UV}}} = \{O_a(x)\}_{a=1}^{N_{\mathrm{con}}},$$

with support radius $O(\ell_{\mathrm{UV}})$ and with label set independent of the number of regulator cells. Under refinement, the same finite constraint family persists, so the realized states belong to one common finite-dimensional MaxEnt family rather than unrelated maximizers.

Axiom 5.4 (Recoverable Generalized Entropy)**.** *A generalized entropy functional exists on caps,*

$$S_{\mathrm{gen}}(C) = S_{\mathrm{bulk}}(C) + \langle L_C \rangle,$$

where L_C is a positive edge-center entropy functional and the semiclassical branch identifies its leading coarse-grained contribution with $A(\partial C)/(4G)$. The functional obeys the recoverability and focusing structure required by the collar and null-modular arguments.

Axiom 5.5 (Minimal Admissible Realization)**.** *Among admissible realized low-energy sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the EFT branch, its admissible light chiral matter content, and one Higgs doublet, the realized package is lexicographically minimal under*

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}},\, N_{\mathrm{nonab}},\, N_c,\, N_g),$$

subject to loop coherence, anomaly freedom, refinement-stable light chiral matter, single-Higgs Yukawa completable structure with one connected abelian charge factor, intrinsic CP capability, and weak-sector UV completeness.

For particle physics, the first four axioms provide the observer-centric kinematic and entropic/modular background. The fifth axiom, MAR, is the selector that turns "some compact gauge group reconstructed from the transportable edge-sector category under the compact paper's explicit colimit and fiber-functor premises" into the realized Standard Model branch. This paper therefore uses MAR constantly, but only after the compact-gauge reconstruction step has been separated from the later realized-branch selection.

5.2 Regulated screen architecture and patch language

The SM/GR derivation in Ref. [4] states the axioms in algebraic language. Ref. [3] supplies one concrete regulated realization of that language: a cellulated horizon screen, finite link or quantum-link registers on the edges, record registers on selected vertices or coarse cells, boundary-fixed patch algebras, overlap algebras on collars, typed readout packets, and local repair instruments. This regulated architecture is important for the particle derivation for three reasons.

First, it makes the screen picture operational. A patch is a finite local algebra carried by one region of the cellulated screen, and an overlap is a boundary-visible algebra with declared shared observables. Second, it makes the edge-sector language concrete. The particle derivation ultimately uses edge sectors, transport, refinement, fusion, and overlap data to build the gauge and flavor branches; Ref. [3] shows how those objects can be realized at fixed cutoff by finite gauge-aware registers. Third, it clarifies the measurement interface. The observer-facing measurement package is not an added Copenhagen layer but a theorem-bearing fixed-cutoff statement about a central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting algebra.

This regulated architecture should be read as a reference implementation, not as the unique OPH UV completion. Ref. [3] is explicit on that boundary: it selects one finite gauge-register or quantum-link style screen model because it is local, finite-dimensional, gauge-aware, and naturally compatible with collars, overlap observables, and repair maps. This paper inherits that architecture as a concrete screen-side realization of the OPH language, while leaving uniqueness and continuum lift open.

5.3 External inputs and where they enter the particle derivation

The quantitative derivation developed here uses two external continuous inputs:

$$P \equiv a_{\mathrm{cell}}/\ell_P^2 = 1.63094, \tag{1}$$

$$N_{\mathrm{scr}} \equiv \log \dim \mathcal{H}_{\mathrm{tot}} \sim 10^{122}. \tag{2}$$

The particle-spectrum draft needs to keep these inputs visible because several later sections depend on them in different ways.

Pixel area P. The pixel area is the declared local UV-area input. In this derivation it feeds the heat-kernel / edge-law input stage and, through that route, the forward D10 electroweak surface. It is the common upstream numerical input for the electroweak, Higgs/top, flavor, quark, and hadron-facing branches. The particle paper therefore treats P as a declared input to the quantitative derivation, not as a quantity derived inside the particle-spectrum argument itself.

Screen capacity N_{scr}. The screen capacity is not part of the local recovered-core gravity/gauge derivation. It enters the separate screen-capacity branch once the screen-capacity identification is adopted. In the particle context, this matters mainly for a legacy capacity-level neutrino side estimate and for the cosmological-capacity discussion that frames why local null data do not determine the cosmological constant. This paper should therefore keep the neutrino chapter direct: legacy capacity-level order-of-magnitude bookkeeping and the live weighted-cycle neutrino derivation are not the same thing.

Why these are the only declared continuous inputs. One of the discipline constraints of the OPH derivation is that particle-sector freedom should not be hidden in a large uncontrolled parameter set. This paper compresses the continuous external freedom into P and N_{scr}, then asks the overlap, modular, gauge, and admissibility machinery to do the rest. Whether later continuation branches remain faithful to that discipline is one of the central questions this paper is meant to make explicit. For the particle-spectrum branch specifically, the nontrivial common quantitative burden sits on P: N_{scr} enters the separate capacity/cosmology side, whereas the reverse-engineering claim for masses and couplings is that one shared pixel constant should drive the spectrum instead of a long sector-by-sector input list.

5.4 Technical premises carried into the particle-spectrum derivation

This paper uses the same technical-premise boundary as *Recovering Relativity and Standard Model Structure from Observer Overlap Consistency* [4], together with the regulator-language clarification integrated into the consensus and microphysics appendices. The premise ledger matters only insofar as it separates branch-internal statements from theorem-external ones.

The third axiom internalizes several pieces of infrastructure that often appear as separate assumptions: the local Gibbs form of the regulator-scale state, the quasi-local propagation bound, the endpoint-control estimate for bounded intervals, and the meaning of refinement stability itself. In regulator language, one works with finite local Hilbert spaces and a boundary-fixed overlap action on cut data, often summarized as the R0/R1 presentation. Beyond that regulator package, the extra burdens are the controlled scaling-limit scope, fixed-cap stationarity in the Jacobson step, the bosonic and fiber-functor clauses required by compact-gauge reconstruction, and the vanishing gluing obstruction when global transportability is invoked. Any local-Gibbs or collar-mixing language on that branch is therefore only fixed-cutoff recoverability/support control; it is not what

decides whether the refinement-limit gauge-sector colimit exists or whether, if supplied, it is trivial or nontrivial. For the BW/geometric side, the target is the extracted prime geometric subnet, and the smallest internal extension route is a two-step chain: first a scaling-limit geometric cap-pair extraction theorem on that subnet that produces transported cap limits with vanishing carried collar errors, and then an ordered null cut-pair rigidity theorem that collapses the residual cap-preserving conformal freedom on those extracted cap pairs to the hyperbolic subgroup. Beneath that carried-collar witness, the sharper raw datum is a fixed-local-collar Markov/faithfulness package: collarwise conditional mutual information vanishing together with an eventual common floor on the finite modular-transport family on each fixed local collar model. Within that datum the single missing clause is precisely that modular-transport common floor, which feeds the faithful modular-defect term. No separate exact-Markov-reference faithfulness input is missing there: on one fixed collar model, the exact-Markov comparison marginals inherit the same eventual floor once the exact-Markov modulus tends to zero.

For the particle-spectrum derivation, this has a practical consequence. Structural statements such as compact-gauge reconstruction, the realized Standard Model quotient, the exact hypercharge lattice, the generation count $N_g = 3$, the color count $N_c = 3$, and the symmetry-protected massless carrier zeros are not floating independently of the OPH premise map. They live on a controlled chain whose scope conditions need to remain visible in this paper, especially once later chapters turn from structural branches to calibration, continuation, and nonperturbative sectors.

5.5 Core theorem packages used later

This paper will repeatedly use earlier OPH results without re-proving them in full. The theorem packages imported into the present draft are the following.

1. **Patch-net and overlap language.** Ref. [2] organizes overlap repair, fixed-point language, cycle holonomy, and gauge-as-gluing in a form that is especially useful for flavor transport and observer-centric interpretation.

2. **Screen-side regulated architecture.** Ref. [3] gives the finite gauge-register screen model, the regulated patch-net embedding theorem, and the fixed-cutoff edge heat-kernel / Casimir theorem.

3. **Measurement interface.** Ref. [3], together with the integrated measurement appendices and the synthesis paper *Observers Are All You Need* [1], supplies the fixed-cutoff central-record, Born-rule, and Lüders-conditioning package used when later chapters discuss how measurements pick out definite observer-accessible outcomes.

4. **Compact gauge reconstruction and Standard Model structure.** *Recovering Relativity and Standard Model Structure from Observer Overlap Consistency* [4] and the dedicated gauge-group fragment together provide the path from refinement-stable edge sectors to a compact group, then from MAR to

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3.$$

5. **Continuation-level topic notes.** The integrated lane appendices carry useful material on supersymmetry, baryogenesis, proton stability, generation structure, and Yukawa hierarchy, but those sections have to be read with their stated status boundaries intact. They are sources for later discussion and continuation chapters, not automatic upgrades to recovered-core theorems.

This reading rule will govern the whole paper. Structural gauge and carrier results can be used as structural results. The D10 electroweak branch can be used as a forward-emitted Phase-II quantitative branch. The Higgs/top critical stage can be used as a secondary quantitative stage. The quark, charged-lepton, neutrino, and hadron chapters retain their stated continuation or simulation status unless the underlying derivation chains close more strongly.

6 Gauge Architecture, Standard Model Structure, and Structural Carriers

The particle-spectrum derivation does not start from a list of particles. It starts from the compact-gauge branch. OPH first reconstructs a compact gauge group from the transportable refinement-directed edge-sector colimit under the compact paper's explicit transportability, symmetry, and fiber-functor premises and only then applies MAR to select the realized low-energy branch. That logical order matters for this paper. The gauge branch fixes the structural carrier content before any mass readout, and it also explains why some rows are exact structural zeros rather than fitted or accidentally small numbers.

On the realized branch, the gauge/content result used throughout this paper is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3.$$

Recovering Relativity and Standard Model Structure from Observer Overlap Consistency [4], the longer main derivation surface, and the dedicated gauge-group proof fragment all agree on the logical split: compact-gauge reconstruction first, realized Standard Model selection second. The first stage gives a compact internal symmetry group under the stated categorical premises. The second stage uses MAR, anomaly cancellation, the one-Higgs admissible matter package, intrinsic CP capability, weak-sector UV completeness, and Witten-parity input to select the realized quotient, the generation count, and the color count.

For the particle zoo this structural branch fixes three exact carrier statements. First, the photon is massless on the realized electromagnetic branch. Second, the gluons are massless on the color branch, even though free gluons are never seen as asymptotic particles because color is confined. Third, the graviton is massless on the dynamical-metric branch of the gravity chain. These are symmetry-protected zeros of the realized branch, not continuation-level masses.

Carrier	Claim tier	OPH output	Physical role	Why the mass is zero
Photon	structural	0 GeV	electromagnetic force carrier	gauge redundancy forbids a hard mass
Gluons	structural	0 GeV	color-force carriers	color-gauge redundancy forbids a hard mass
Graviton	structural	0 GeV	spin-2 metric carrier	diffeomorphism redundancy forbids a hard mass

These structural zeros should not be confused with the later quantitative particle rows. The paper should therefore treat the photon, gluons, and graviton as consequences of the gauge and gravity branches themselves, not as outputs of the later calibration or flavor machinery. The later boson sections will add the W boson, the Z boson, and the Higgs boson, but those are not structural zeros; they sit on electroweak and Higgs/top quantitative stages.

7 Electroweak D10 Branch and the Higgs/Top Critical Stage

The reported bosonic sector distinguishes sharply between emitted numerical rows and missing rows. The active bosonic stages are the electroweak D10 prediction stage and the Higgs/top critical stage, while charged-lepton rows remain explicit gaps, the neutrino lane closes on a separate weighted-cycle theorem branch, and hadron rows are suppressed. The bosonic numerical sector is therefore concentrated in two places: the D10 electroweak branch for the W boson and Z boson, and the Higgs/top critical stage for the Higgs boson, together with the top quark.

7.1 Electroweak prediction stage on the D10 calibration branch

The electroweak derivation consists of a single-P running family followed by a reduced two-scalar carrier, a selector on that carrier, an exact carrier mass chart, and then a target-free source-only repair theorem beyond the present carrier. In practical terms, the construction starts from the declared pixel input P, builds the running electroweak family, reduces it to the present carrier, reads the W boson/Z boson pair from the selected point on that carrier, and then emits the coherent repaired electroweak quintet from the source basis alone. On the D10 calibration branch, forward prediction means exactly that ordering: first fix the shared pixel input P on the declared D10 running/matching surface, thereby fixing the source basis

$$(\alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v),$$

with $\alpha_U(P)$, equivalently $t_U(P)$ and $t_{\text{tr}}(P)$, fixed by the same forward pixel-closure solve, and only then emit the repaired electroweak quintet forward from that source basis. The active public W/Z rows are therefore not obtained by re-inserting the target masses after calibration; they are forward outputs of the closed D10 Phase-II calibration theorem.

The public reported rows are

$$m_W = 80.37700001539531 \text{ GeV}, \qquad m_Z = 91.18797807794321 \text{ GeV}.$$

These numbers are emitted on the target-free source-only repair theorem, not on the unmodified selected carrier. The exact current-carrier chart remains explicit on disk and emits the local pair

$$m_W^{\text{carrier}} = 80.38629169244275 \text{ GeV}, \qquad m_Z^{\text{carrier}} = 91.18290444674243 \text{ GeV},$$

so the mathematics distinguishes three electroweak objects: the exact current-carrier chart, the freeze-once coherent repair surface, and the target-free source-only theorem surface that supplies the public rows.

Theorem 7.1 (Freeze-once coherent electroweak repair law)**.** *Let the emitted running/core electroweak basis be*

$$Q_{\text{run}}(P) = \big(\alpha_{Y,m_Z}(P), \alpha_{2,m_Z}(P), v(P), \eta_{\text{source}}(P)\big),$$

and write

$$\tau_Y(\tau_2) = -\frac{\tau_2 + 2\eta_{\text{source}}}{1 + 4\tau_2^2}, \qquad n_{\text{fiber}}(\tau_2) = 1 + \frac{\alpha_{Y,m_Z}\tau_Y(\tau_2) + \alpha_{2,m_Z}\tau_2}{\alpha_{Y,m_Z} + \alpha_{2,m_Z}}.$$

Freeze one authoritative target pair

$$\mathcal{T}^\dagger = (M_W^\dagger, M_Z^\dagger), \qquad 0 < M_W^\dagger < M_Z^\dagger,$$

and define the charged anchor

$$\tau_{2,W}^{\dagger} = \frac{M_W^{\dagger\,2}}{\pi v^2 \alpha_{2,m_Z}} - 1, \qquad \delta\alpha_2^{\dagger} = \frac{M_W^{\dagger\,2}}{\pi v^2} - \alpha_{2,m_Z}.$$

Then the fiber-parallel hypercharge leg is

$$\tau_Y^{\dagger} = \tau_Y(\tau_{2,W}^{\dagger}), \qquad n_{\text{fiber}}^{\dagger} = n_{\text{fiber}}(\tau_{2,W}^{\dagger}),$$

$$M_{Z,\text{fiber}}^{\dagger} = v\sqrt{\pi(\alpha_{Y,m_Z} + \alpha_{2,m_Z})\, n_{\text{fiber}}^{\dagger}},$$

and the orthogonal neutral-shear scalar is

$$\delta M_Z^{\perp} = M_Z^{\dagger} - M_{Z,\text{fiber}}^{\dagger},$$

equivalently

$$\delta n^{\dagger} = \frac{(M_Z^{\dagger} + M_{Z,\text{fiber}}^{\dagger})\, \delta M_Z^{\perp}}{\pi v^2(\alpha_{Y,m_Z} + \alpha_{2,m_Z})}, \qquad \delta\alpha_Y^{\perp} = \frac{(M_Z^{\dagger} + M_{Z,\text{fiber}}^{\dagger})\, \delta M_Z^{\perp}}{\pi v^2}.$$

The fiber-parallel hypercharge motion is

$$\delta\alpha_Y^{\parallel} = \alpha_{Y,m_Z}\, \frac{8\eta_{\text{source}}(\tau_{2,W}^{\dagger})^2 - \tau_{2,W}^{\dagger}}{1 + 4(\tau_{2,W}^{\dagger})^2}.$$

Therefore the unique coherent repair package is

$$\Sigma_{EW}^{\dagger} = (\delta\alpha_2^{\dagger}, \delta\alpha_Y^{\parallel}, \delta\alpha_Y^{\perp}),$$

equivalently $(\tau_{2,W}^{\dagger}, \delta n^{\dagger})$*. If the selected carrier anchor is the* $\tau_2 = 0$ *fiber point, so that*

$$\alpha_{2,*} = \alpha_{2,m_Z}, \qquad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\text{source}}),$$

then the repaired coherent couplings are

$$\alpha_{2,\dagger} = \alpha_{2,*} + \delta\alpha_2^{\dagger}, \qquad \alpha_{Y,\dagger} = \alpha_{Y,*} + \delta\alpha_Y^{\parallel} + \delta\alpha_Y^{\perp},$$

and they emit one coherent quintet

$$M_W^{\dagger} = v\sqrt{\pi\alpha_{2,\dagger}}, \qquad M_Z^{\dagger} = v\sqrt{\pi(\alpha_{Y,\dagger} + \alpha_{2,\dagger})},$$

$$\alpha_{\text{em},\dagger}^{-1} = \frac{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}{\alpha_{Y,\dagger}\alpha_{2,\dagger}}, \qquad \sin^2\theta_{W,\dagger} = \frac{\alpha_{Y,\dagger}}{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}.$$

So the frozen-target electroweak repair is not another chart search or selector shell. It is one unique coherent value-emission law on top of the closed fiber map.

For the local PDG-2025 calibration surface,

$$\begin{aligned} \alpha_{2,m_Z} &= 0.03377843630219015, & v &= 246.76711732749683 \text{ GeV}, \\ \alpha_{Y,m_Z} &= 0.010131601067241624, & M_W^{\dagger} &= 80.377 \text{ GeV}, \\ \eta_{\text{source}} &= 0.022147000871961295, & M_Z^{\dagger} &= 91.18797809193725 \text{ GeV}, \end{aligned}$$

and the law emits

$$\begin{aligned}
\tau_{2,W}^{\dagger} &= -2.3116268316325517 \times 10^{-4}, \\
\delta\alpha_2^{\dagger} &= -7.808313968675484 \times 10^{-6}, \\
M_{Z,\mathrm{fiber}}^{\dagger} &= 91.17717018044121 \text{ GeV}, \\
\delta M_Z^{\perp} &= 10.807911496044653 \text{ MeV}, \\
\delta\alpha_Y^{\parallel} &= 2.3421435088707883 \times 10^{-6}, \\
\delta\alpha_Y^{\perp} &= 1.0302892666790446 \times 10^{-5}, \\
\delta n^{\dagger} &= 2.3463639031113336 \times 10^{-4}.
\end{aligned}$$

So the repaired coherent couplings are

$$\alpha_{2,\dagger} = 0.03377062798822147, \qquad \alpha_{Y,\dagger} = 0.00969547694807616,$$

and the same coherent branch emits

$$M_W^{\dagger} = 80.377 \text{ GeV}, \qquad M_Z^{\dagger} = 91.18797809193725 \text{ GeV},$$

$$\alpha_{\mathrm{em},\dagger}^{-1} = 132.752408550967, \qquad \sin^2\theta_{W,\dagger} = 0.223058333896849.$$

That freeze-once law provides an exact coherent validation surface. On that frozen authoritative repair surface the canonical W/Z pair is hit exactly, but only as compare-only validation. The active public electroweak surface is the target-free source-only repair law.

Theorem 7.2 (Target-free source-only electroweak repair law)**.** *Let the D10 source-only emitted basis be*

$$Q_{\mathrm{src}}(P) = \left(\alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\mathrm{source}}(P), v(P)\right),$$

and define

$$\rho_{\mathrm{EW}} := \frac{\alpha_{2,m_Z} - \alpha_{Y,m_Z}}{\alpha_{2,m_Z} + \alpha_{Y,m_Z}}, \qquad \alpha_u^{\mathrm{seed}} := \frac{\eta_{\mathrm{source}}}{\rho_{\mathrm{EW}}}, \qquad \lambda_{\mathrm{EW}} := \frac{\eta_{\mathrm{source}}\alpha_u^{\mathrm{seed}}}{4} = \frac{\eta_{\mathrm{source}}^2}{4\rho_{\mathrm{EW}}}.$$

Then the beyond-current-carrier repair is uniquely

$$\tau_2^{\mathrm{exact}} = -\lambda_{\mathrm{EW}}\left(1 + \frac{2}{3}\eta_{\mathrm{source}} + \left(1 - \frac{\rho_{\mathrm{EW}}}{6}\right)\eta_{\mathrm{source}}^2\right),$$

$$\delta n^{\mathrm{exact}} = \lambda_{\mathrm{EW}}\left(1 + \frac{4}{3}\eta_{\mathrm{source}} + \left(2 - \frac{\rho_{\mathrm{EW}}}{6}\right)\eta_{\mathrm{source}}^2\right),$$

with fiber hypercharge transport

$$\tau_Y^{\mathrm{fiber}} = -\frac{\tau_2^{\mathrm{exact}} + 2\eta_{\mathrm{source}}}{1 + 4(\tau_2^{\mathrm{exact}})^2}.$$

The repaired coupling shifts are

$$\delta\alpha_2 = \alpha_{2,m_Z}\tau_2^{\mathrm{exact}}, \qquad \delta\alpha_Y^{\parallel} = \alpha_{Y,m_Z}\frac{8\eta_{\mathrm{source}}(\tau_2^{\mathrm{exact}})^2 - \tau_2^{\mathrm{exact}}}{1 + 4(\tau_2^{\mathrm{exact}})^2},$$

$$\delta\alpha_Y^{\perp} = \left(\alpha_{2,m_Z} + \alpha_{Y,m_Z}\right)\delta n^{\mathrm{exact}}.$$

With

$$\alpha_{2,*} = \alpha_{2,m_Z}, \qquad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\text{source}}),$$

the repaired couplings are

$$\alpha_2' = \alpha_{2,*} + \delta\alpha_2, \qquad \alpha_Y' = \alpha_{Y,*} + \delta\alpha_Y^{\parallel} + \delta\alpha_Y^{\perp},$$

and they emit one coherent family

$$W = v\sqrt{\pi\alpha_2'}, \qquad Z = v\sqrt{\pi(\alpha_2' + \alpha_Y')},$$

$$\alpha_{\text{em}}^{-1} = \frac{\alpha_2' + \alpha_Y'}{\alpha_2'\alpha_Y'}, \qquad \sin^2\theta_W = \frac{\alpha_Y'}{\alpha_2' + \alpha_Y'}, \qquad v = v(P).$$

So the electroweak repair is emitted from the D10 source basis alone, with no frozen target input, no inverse slice, and no mixed readout.

Using the D10 source basis

$$\begin{aligned} \alpha_{2,m_Z} &= 0.03377843630219015, \\ \alpha_{Y,m_Z} &= 0.010131601067241624, \\ \eta_{\text{source}} &= 0.022147000871961295, \\ v &= 246.76711732749683 \text{ GeV}, \end{aligned}$$

the theorem emits

$$W = 80.37700001539531 \text{ GeV}, \qquad Z = 91.18797807794321 \text{ GeV},$$

$$\alpha_{\text{em}}^{-1} = 132.7524088191668, \qquad \sin^2\theta_W = 0.22305833336075578.$$

The freeze-once coherent repair surface provides compare-only validation, with machine-scale deltas

$$\Delta M_W \approx 1.54 \times 10^{-8} \text{ GeV}, \qquad \Delta M_Z \approx -1.40 \times 10^{-8} \text{ GeV}.$$

The electroweak calibration branch also contains several subordinate theorem objects: an underdetermination theorem for the quadratic repair family, the smallest conditional route through $\text{ColorBalancedQuadraticRepairDescent}_{EW}$, and the source-only expression $\text{EWTargetEmitter}_{D10}$ whose promotion yields the active theorem. A further forward transmutation certificate sits directly beneath the promoted readout and records that the source-only basis reconstructs the same $\alpha_U(P)$, $t_U(P)$, and $t_{\text{tr}}(P)$ as the pixel-closure solve without reading them back from measured couplings. These objects locate the public repair law inside the calibration branch without changing the emitted public surface.

7.2 Higgs/top critical stage

The Higgs/top critical stage is a secondary quantitative stage built on top of the electroweak gauge core. It is described by a compact shared seed together with a Jacobian readout map. The reported rows use a promoted forward seed whose certificate closes the fixed-ray forward path on its emitted one-scalar branch. This gives

$$m_H = 125.218922 \text{ GeV}, \qquad m_t = 172.388646 \text{ GeV}.$$

On this branch the paper-facing closure contract is one-scalar: one emitted D11 forward seed determines one forward Higgs/top pair on the fixed ray. The forward seed is closed on that emitted branch, so the public rows are substantive D11 outputs of the critical stage. Separately, the same D11 Jacobian admits a compare-only inverse slice that hits the canonical Higgs/top reference pair exactly. That exact sidecar is useful for bookkeeping and does not define the forward branch used for the public rows. The Higgs boson belongs naturally in the boson discussion, while the top quark will be revisited in the quark-family chapter as the third-generation up-type quark. The reported top-quark row is carried by the Higgs/top critical stage, not by the quark derivation itself.

The distinction is between the closed forward seed and the broader paper surface. The forward seed is closed on its emitted one-scalar branch, so the Higgs/top rows are substantive emitted outputs. The current paper-level boundary is presentation-level rigor and synchronization across the broader OPH exposition, not the existence of a larger missing mass-moving coordinate on this branch.

Particle	Stage	Claim tier	Reported value
W boson	D10 electroweak branch	calibration	80.37700001539531 GeV
Z boson	D10 electroweak branch	calibration	91.18797807794321 GeV
Higgs boson	Higgs/top critical stage	secondary quantitative	125.218922 GeV
Top quark	Higgs/top critical stage	secondary quantitative	172.388646 GeV

The electroweak chapter therefore presents three mathematically distinct surfaces: the reported target-free source-only theorem rows, the current-carrier chart beneath them, and the freeze-once coherent repair surface used for compare-only validation. That is the technically direct way to present the bosonic sector.

8 Flavor Transport, Generation Structure, and the Yukawa Dictionary

The gauge branch explains why the realized low-energy sector has three generations, but it does not yet by itself produce the full flavor dictionary. This paper therefore needs a separate flavor chapter. The imported realized-branch premise here is the D9 counting chain $N_g = 3$ then $N_c = 3$. This chapter is the bridge between that structural result and the matter-family chapters for quarks, charged leptons, and neutrinos.

The flavor derivation is deliberately constructive. It does not claim that the full OPH flavor observable is theorem-level. Instead, it builds the object chain that a closed theorem would have to pass through. The derivation chain has the following main architecture: start from a refinement-indexed family transport kernel, derive a centered generation-bundle branch generator, lift that to same-label transport data, derive the induced overlap-edge transport cocycle, reduce the cocycle to a persistent flavor observable, and then push that observable into common sector-response objects for the downstream quark, charged-lepton, and neutrino derivations.

Mathematically, this derivation is where "why this family exists" becomes a concrete technical question. The realized Standard Model branch gives three generations structurally; the flavor derivation is where one tries to turn that three-generation existence statement into transport, splitting, suppression, phase, and excitation data. The relevant objects are not yet the final fermion masses themselves but the intermediate transport and spectral structures that later mass readouts consume.

The active chain, in slightly compressed form, is:

1. normalize a refinement-indexed family transport kernel;
2. derive a centered generation-bundle branch generator on the realized three-generation charged bundle;
3. lift this to same-label edge-line transport and then to an overlap-edge cocycle with explicit defect and gap bookkeeping;
4. reduce those data to projectors, spectral gaps, pair suppressions, and cycle phases;
5. push the resulting family object into sector-response objects and then into suppression/phase tensors for the downstream matter derivations.

Three lane-specific claim boundaries are attached to this region of the derivation. The shared excitation dictionary is the common proof-facing base. Above that base, the charged lane carries exact centered readback and a closed common-shift no-go, then waits on promotion of $\widehat{C}_e^{\mathrm{cand}}$ and the post-promotion descended scalar $\mu_{\mathrm{phys}}(Y_e)$, the neutrino lane closes on the weighted-cycle bridge-rigidity surface above the emitted proxy P_ν, with induced paper-facing amplitude B_ν and absolute attachment above the scale-free branch, and the quark lane carries the internalized mass bridge together with the selected-class public theorem on the public physical quark frame class f_P. That theorem makes the exact physical sigma datum target-free public on f_P, emits (g_u, g_d) algebraically through the affine mean law, and emits the exact running sextet together with explicit exact forward Yukawas Y_u and Y_d. A separate compare-only microphysical bridge records edge-statistics transport on a diagnostic surface. Together these boundaries mark where the flavor derivation leaves the common transport backbone and passes to lane-specific closure contracts.

9 Quark Family Derivation

The quark lane carries a theorem-grade selected-class closure on the public physical quark frame class f_P chosen by P. On that selected class the exact physical sigma datum is target-free public, the affine mean law emits (g_u, g_d) algebraically, the ordered three-point quadratic readout emits the exact running quark sextet, and the exact forward construction emits explicit exact forward Yukawas Y_u and Y_d. The same sextet is also realized by two supporting exact surfaces: the exact same-family witness on `current_family_only` and the restricted current-family common-refinement transport-frame theorem on `current_family_common_refinement_transport_frame_only`. This is selected-class closure only. It does not classify all public quark frame classes.

9.1 What quarks are in this derivation

Quarks are the color-charged elementary constituents of hadronic matter. In nature, the up quark and down quark dominate protons and neutrons, while the strange quark, charm quark, bottom quark, and top quark appear in progressively heavier and more unstable sectors. In the OPH particle derivation, quarks are the only matter family whose flavor lane carries an exact theorem-grade running-mass readout together with explicit exact forward Yukawas on a public selected class.

The quark route starts from the shared flavor excitation dictionary, internalizes the target-free mass bridge on the D12 mass ray, descends the exact physical sigma datum to the selected public class f_P, applies the affine mean law, and closes both the ordered three-point readout and the exact forward construction. The historical same-label left-handed selector surface closes to the singleton

σ_{ref}. That lower object is a negative sheet-selector statement on the selected D12 sheet. It does not replace the selected-class public theorem.

9.2 Emitted quark rows

Quark	Claim tier	Exact running value	Derivation stage
Up quark	selected-class theorem	0.00216 GeV	selected public quark frame class f_P
Down quark	selected-class theorem	0.00470 GeV	selected public quark frame class f_P
Strange quark	selected-class theorem	0.0935 GeV	selected public quark frame class f_P
Charm quark	selected-class theorem	1.273 GeV	selected public quark frame class f_P
Bottom quark	selected-class theorem	4.183 GeV	selected public quark frame class f_P
Top quark	selected-class theorem	172.3523553288311 GeV	selected public quark frame class f_P

These comparisons are made against running quark references, not direct free-particle pole masses, because quarks are confined. That is not a technicality; it is part of what the quark chapter must explain to technical readers. The exact sextet matches the official PDG 2025 API running-quark target surface, with the top coordinate taken from the PDG cross-section entry rather than the auxiliary direct-top entry. The Higgs/top branch also carries its own public top row 172.388646 GeV on the D11 forward seed. The table above records the quark-family running-mass theorem surface on f_P.

9.3 Target-free mass bridge and selected-class public theorem

The historical same-label left-handed solver surface closes to the singleton σ_{ref} (`sigma_ref`). This is a negative same-sheet selector statement: same-sheet rephasing preserves CKM moduli, so it does not move that selected sheet to the physical CKM shell. Same-sheet overlap scans, chirality-swapped basis diagnostics, and other non-sector-attached orbit improvements are compare-only and do not change the selected-class theorem.

A separate target-free mass bridge is internalized on the emitted D12 ray. On the minimal light branch

$$y_u = c_u\,\varepsilon^6, \qquad y_d = c_d\,\varepsilon^6, \qquad \varepsilon = \frac{1}{6},$$

the light-quark overlap-defect theorem emits

$$\Delta_{ud}^{\mathrm{overlap}} = \frac{1}{6}\log\frac{c_d}{c_u}.$$

The emitted same-family D12 mass object is the ray

$$D12_{ud}^{\mathrm{mass}} \equiv \mathcal{R}_{D12}^{ud} \qquad \text{(paper id } \texttt{D12_ud_mass_ray}\text{)},$$

and on that ray the same scalar is equivalently the one-scalar law

$$\Theta_{ud}^{\mathrm{mass}} := \texttt{quark_d12_t1_value_law}.$$

The exact scalar identities are

$$\Delta_{ud}^{\mathrm{overlap}} = \frac{t_1}{5}, \qquad \log\frac{c_d}{c_u} = \frac{6}{5}\,t_1, \qquad t_1 = 5\,\Delta_{ud}^{\mathrm{overlap}} = \frac{5}{6}\log\frac{c_d}{c_u}.$$

The D12 mass-side package is therefore functorial:

$$\eta_Q^{\text{centered}} = -\frac{1-x_2^2}{27}\,t_1, \qquad \kappa_Q = -\frac{t_1}{54}, \qquad x_2 = -0.5175863354681689.$$

The odd source package is also forced:

$$\beta^{\text{source}}_{u,\text{diag},B} = \frac{t_1}{10},$$
$$\beta^{\text{source}}_{d,\text{diag},B} = -\frac{t_1}{10},$$

$$\texttt{source_readback_u_log_per_side} = \left(-\frac{t_1}{10},\, 0,\, +\frac{t_1}{10}\right),$$
$$\texttt{source_readback_d_log_per_side} = \left(+\frac{t_1}{10},\, 0,\, -\frac{t_1}{10}\right),$$

and with the emitted spread pair

$$\sigma_u = 5.5905, \qquad \sigma_d = 3.3049,$$

one gets

$$\tau_u = \frac{\sigma_d}{10(\sigma_u+\sigma_d)}\,t_1, \qquad \tau_d = \frac{\sigma_u}{10(\sigma_u+\sigma_d)}\,t_1.$$

The mass bridge is not part of the selected-class theorem boundary. The builder-facing pure-B payload pair is a lower implementation object beneath the exact theorem surface, and `intrinsic_scale_law_D12` is the derived wrapper above $\Theta^{\text{mass}}_{ud}$.

Write the same-label left-handed physical carrier as

$$\Sigma^{\text{phys}}_{ud} := \left\{(\sigma_{\text{id}}, \tau, U_{u,L}, U_{d,L}, V_{\text{CKM}}, I_{\text{CKM}}) : V_{\text{CKM}} = U^\dagger_{u,L}U_{d,L}\right\}/\sim,$$

where

$$(U_{u,L}, U_{d,L}, V) \sim (U_{u,L}D_u,\ U_{d,L}D_d,\ D_u^\dagger V D_d)$$

for diagonal $D_u, D_d \in U(1)^3$. On the selected public quark frame class f_P, represented on the realized lane by the explicit common-refinement transport-frame class $[F_0^\dagger F_1]$, the exact $\Sigma^{\text{phys}}_{ud}$ element and the attached exact sigma datum are independent of the declared representative inside the selected bridge fiber. Therefore the exact sigma readout descends uniquely to target-free public data on f_P.

Define

$$\sigma^{ud}_{\text{seed}} := \frac{\sigma_u+\sigma_d}{2}, \qquad \eta_{ud} := \frac{\sigma_u-\sigma_d}{2},$$

$$A_{ud} := \frac{1}{2(1+\rho_{\text{ord}}-x_2^2)}, \qquad B_{ud} := \frac{1}{2\left(1-x_2^2-\frac{x_2^2}{1+\rho_{\text{ord}}}\right)}, \qquad \rho_{\text{ord}} = 1.294284936377706.$$

Then

$$g_u = g_{\text{ch}}\exp\!\big(-(A_{ud}\sigma^{ud}_{\text{seed}} - B_{ud}\eta_{ud})\big), \qquad g_d = g_{\text{ch}}\exp\!\big(-(A_{ud}\sigma^{ud}_{\text{seed}} + B_{ud}\eta_{ud})\big).$$

The affine mean law therefore emits (g_u, g_d) algebraically on f_P. The ordered three-point quadratic readout emits the exact running quark sextet, and the exact forward construction emits explicit exact forward Yukawas Y_u and Y_d on that selected public class. The selected-class public theorem can be written as

$$f_P \Longrightarrow (\sigma_u, \sigma_d, \sigma^{ud}_{\text{seed}}, \eta_{ud})_{\text{phys}} \Longrightarrow (g_u, g_d) \Longrightarrow (m_u, m_d, m_s, m_c, m_b, m_t),$$

together with explicit exact forward Yukawas Y_u and Y_d on the same selected-class surface.

Maximal theorem-emitted quark package theorem. Let

$$\begin{aligned} P := p_a & \\ & + (\text{Axioms 1–5}) \\ & + \text{Assumption 6} \\ & + \text{the emitted OPH nodes D1–D10} \\ & + \text{the listed public D12 quark objects.} \end{aligned}$$

Then the quark-side package can be written in four layers:

1. the emitted D12 mass ray

$$D12_{ud}^{\text{mass}} = \mathcal{R}_{D12}^{ud} = \left\{ \lambda \left(\frac{1}{5}, -\frac{1 - x_2^2}{27} \right) : \lambda \geq 0 \right\}, \qquad \lambda = \text{ray_modulus} = t_1;$$

2. the historical same-label left-handed selector value

$$\sigma_{ud} = \sigma_{\text{ref}},$$

   with canonical token `D12::same_label_left::reference_sheet`, which is a negative sheet-selector statement;

3. the separate target-free mass bridge

$$P \vdash \Delta_{ud}^{\text{overlap}} = \frac{1}{6} \log \frac{c_d}{c_u}, \qquad P \vdash \Theta_{ud}^{\text{mass}} = \texttt{quark_d12_t1_value_law};$$

4. the selected-class public theorem on f_P, which emits the target-free exact physical sigma datum, the affine pair (g_u, g_d), the exact running sextet, and explicit exact forward Yukawas Y_u and Y_d.

A separate exact same-family witness reproduces the same sextet

$$(u, d, s, c, b, t) = (0.00216, 0.00470, 0.0935, 1.273, 4.183, 172.3523553288311)\,\text{GeV}$$

on `current_family_only`, and the restricted current-family common-refinement transport-frame theorem reconstructs the same sextet and explicit exact forward Yukawas on `current_family_common_refinement_transport_frame_only`. Those supporting surfaces agree with the selected-class public theorem on the exact running target surface.

A separate D12 internal backread sidecar is on disk. On that continuation-only surface, feeding the emitted reference-free forward light-quark pair through the coefficient-ratio identification $c_u/c_d = y_u/y_d = m_u/m_d$ and the D12 normalization $\Delta_{ud}^{\text{overlap}} = \frac{1}{6}\log(c_d/c_u)$ fixes

$$\Delta_{ud}^{\text{overlap}} = 0.13431740757663183, \qquad t_1 = 0.6715870378831591,$$

$$\eta_Q^{\text{centered}} = -0.018210067243314802, \qquad \kappa_Q = -0.012436796997836279.$$

This closes the D12 mass-side scalar package numerically on that sidecar surface. It does not replace the selected-class public theorem and it does not classify all public quark frame classes.

9.4 Quark Theorem Boundary

The exact closure route is

$$f_P \Longrightarrow (\sigma_u, \sigma_d, \sigma^{ud}_{\text{seed}}, \eta_{ud})_{\text{phys}} \Longrightarrow (g_u, g_d) \Longrightarrow (m_u, m_d, m_s, m_c, m_b, m_t),$$

with the exact forward construction emitting Y_u and Y_d on the same selected-class surface. This is selected-class closure only. It does not classify all public quark frame classes.

10 Charged-Lepton Family Derivation

The charged-lepton derivation is one step less mature than the quark derivation. It contains an ordered package, a support-obstruction certificate, a support-extension emitter shell, a two-scalar completion-law shell, the smaller eta source-readback primitive beneath that shell, and a forward shape/scale surface. But unlike the quark derivation, it does not yet emit numerical masses. The electron, muon, and tau rows are therefore reported as n/a.

The mathematical reason is clear. The charged-lepton derivation starts from the ordered charged package, proves that the realized support is a one-dimensional linear subray, exposes the canonical quadratic support-extension direction, maps that into the charged excitation gaps, closes a two-scalar support-extension law shell, isolates the smaller eta source-readback primitive on that same carrier, and then builds the log-spectrum and forward shape/scale surface. The unresolved scalar order is the pair `eta_source_support_extension_log_per_side` and `sigma_source_support_extension_total_log_per_side`. Until those objects are properly closed, the charged-lepton chapter remains a continuation chapter rather than a quantitative one.

Physically, the electron is the light charged lepton that makes atoms and chemistry possible, the muon is its heavier unstable cousin, and the tau lepton is the heaviest charged lepton. The role of the charged-lepton chapter is to explain how those three rows emerge from the shared flavor dictionary without falling back to Koide-assisted fitting. The exact burden is sharper than a generic "emit η and σ" instruction: the theorem-grade lane has to derive promotion of the latent charged sector-response candidate $\widehat{C}_e^{\text{cand}}$ to theorem-grade $\widehat{C}_e$ by closing the branch-generator splitting theorem. Conditional on that promotion, the post-promotion obstruction is smaller than the determinant-line summary suggested: once the uncentered lift is refinement-stable on theorem-grade physical Y_e, the determinant-line section and affine anchor collapse to one descended physical affine scalar $\mu_{\text{phys}}(Y_e)$, with

$$\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\text{phys}}(Y_e)\,\mathbf{1}, \qquad s_{\text{det}}(Y_e) = 3\mu_{\text{phys}}(Y_e), \qquad A_{\text{ch}}(Y_e) = \mu_{\text{phys}}(Y_e).$$

Theorem 10.1 (Charged same-carrier source-pair readback)**.** *Fix the ordered charged carrier*

$$(-1, x_2, 1), \qquad x_2 = -0.5175863354681689.$$

If the charged source pair

$$(\eta_{\text{ext}}, \sigma_{\text{ext}}) = (\eta_{\text{source_support_extension_log_per_side}}, \sigma_{\text{source_support_extension_total_log_per_side}})$$

is emitted on that carrier, then the centered charged logs are

$$e_{\text{log,centered}} = -\frac{(3 + x_2)\sigma_{\text{ext}} - \eta_{\text{ext}}}{6},$$

$$\mu_{\text{log,centered}} = \frac{x_2\sigma_{\text{ext}} - \eta_{\text{ext}}}{3},$$

$$\tau_{\log,\mathrm{centered}} = \frac{(3-x_2)\sigma_{\mathrm{ext}} + \eta_{\mathrm{ext}}}{6},$$

and therefore the charged masses are

$$m_e = g_e\, e^{e_{\log,\mathrm{centered}}}, \qquad m_\mu = g_e\, e^{\mu_{\log,\mathrm{centered}}}, \qquad m_\tau = g_e\, e^{\tau_{\log,\mathrm{centered}}}.$$

This theorem is exact on the same-carrier shell, but the values themselves are open. The visible scalar order remains

$$\eta_{\mathrm{ext}} \quad \text{then} \quad \sigma_{\mathrm{ext}},$$

and the charged absolute scale g_e remains unresolved. Builder-facing code therefore exposes η_{ext} and then σ_{ext} as the first same-carrier residuals, but the theorem-grade waiting set is sharper than that surface suggests. If the latent candidate $\widehat{C}_e^{\mathrm{cand}}$ is promoted, then η_{ext} and σ_{ext} become charged spectral invariants rather than separate primitive goals, and the absolute-scale burden is pushed to one affine-covariant absolute charged anchor A_{ch}. In the local chain, $\widehat{C}_e$ itself is undeclared: only the centered compressed generation-bundle branch-operator candidate $\widehat{C}_e^{\mathrm{cand}}$ is on disk, and the remaining burden for that package is the upstream promotion theorem `oph_generation_bundle_branch_generator_splitting`, not a new ad hoc charged operator ansatz. Its smallest remaining clause is the compression-descendant commutator statement `compression_descendant_commutator_vanishes_or_is_uniformly_quadratic_small_after_central_split`. The centered common-shift quotient is closed negatively, so centered data alone do not emit the affine anchor A_{ch}.

Theorem 10.2 (Charged absolute-scale underdetermination)**.** *Let*

$$E_e^{\mathrm{centered}} = \big(e_{\log,\mathrm{centered}}, \mu_{\log,\mathrm{centered}}, \tau_{\log,\mathrm{centered}}\big)$$

be the centered charged log triple emitted from the charged source pair $(\eta_{\mathrm{ext}}, \sigma_{\mathrm{ext}})$*. Then for every* $c \in \mathbb{R}$*,*

$$Y_e(c) := \exp(c)\,\mathrm{diag}\big(e^{E_e^{\mathrm{centered}}}\big)$$

has the same charged spectral invariants

$$\eta_{\mathrm{ext}}, \qquad \sigma_{\mathrm{ext}}, \qquad \gamma_{21}, \qquad \gamma_{32},$$

the same centered log vector, and the same ratio data. Only the absolute masses scale:

$$(m_e, m_\mu, m_\tau) \longmapsto e^c\,(m_e, m_\mu, m_\tau).$$

Hence the charged OPH chain determines only the quotient class

$$E_e^{\mathrm{centered}} \in \mathbb{R}^3/\langle(1,1,1)\rangle,$$

not the absolute scalar $g_e = e^c$*.*

The proof is immediate from the centered sum rule

$$e_{\log,\mathrm{centered}} + \mu_{\log,\mathrm{centered}} + \tau_{\log,\mathrm{centered}} = 0,$$

which implies

$$\det Y_e^{\mathrm{shape}} = 1, \qquad \det Y_e = g_e^3.$$

All emitted charged invariants depend only on differences of log entries, so a common shift in the $(1,1,1)$ direction leaves them unchanged. This is the exact reason the charged lane remains open: the theorem surface fixes the centered shape, not the absolute normalization.

One important code-level and mathematical cleanup is complete: the charged absolute-scale lane is explicitly typed. The charged scale is a linear quantity

$$g_e = e^{\mu_e^{\mathrm{abs}}},$$

so the log-coordinate μ_e^{abs} must not be mixed directly with centered log gaps. The same-family writeback therefore records the type-consistent shell

$$\mu_{e,\mathrm{seed}}^{\mathrm{abs}} = \log(0.9231656602589082) = -0.07994658034676537,$$

$$\mu_{e,\mathrm{cand}}^{\mathrm{abs}} = \mu_{e,\mathrm{seed}}^{\mathrm{abs}} - \gamma_{\min} = -0.38231224060567365, \qquad g_e^{\mathrm{cand}} = e^{\mu_{e,\mathrm{cand}}^{\mathrm{abs}}} = 0.6822819838027987.$$

This is a representation-consistency shell only, not a charged-mass theorem. It fixes the linear-vs-log coordinate discipline while leaving the true emitted value law for g_e open. The audited shortcut

$$\Delta_e^{\mathrm{abs}} = 0.30236566025890826, \qquad g_e = 0.6822819838027987$$

is not a theorem-grade closure. It merely chooses one representative on the common-shift orbit and does not land on the physical charged masses.

The next theorem object beyond this shell is one theorem-grade affine-covariant section A_{ch} of the quotient map, satisfying

$$A_{\mathrm{ch}}(\ell + c\,\mathbf{1}) = A_{\mathrm{ch}}(\ell) + c.$$

The clean realization of that section is an uncentered charged response lift carrying a determinant line, in which

$$A_{\mathrm{ch}} = \tfrac{1}{3}\log\det(Y_e) = \tfrac{1}{3}\,\mathrm{tr}(\log Y_e).$$

Once such an anchor exists,

$$g_e = e^{A_{\mathrm{ch}}(\ell)}, \qquad \Delta_e^{\mathrm{abs}} = \log g_{\mathrm{ch}}^{\mathrm{shared}} - A_{\mathrm{ch}}(\ell).$$

Compare-only charged continuation bridge. For comparison, there is a sharper charged continuation bridge under three extra continuation assumptions:

1. a uniform $\mathbb{Z}_6$ center-label ensemble, so $\varepsilon = 1/6$;
2. the balanced Hermitian circulant branch, so $|b|/a = 1/\sqrt{2}$;
3. the OPH phase choice $\delta = 2/9$.

With the scale-free normalization $a = 1$, the ordered roots

$$r_k = a + 2|b|\cos\left(\delta + \frac{2\pi k}{3}\right)$$

become

$$r_e = 0.040349908219207475, \qquad r_\mu = 0.5802119201475368, \qquad r_\tau = 2.3794381716332555.$$

Under those assumptions the bridge selects the same-carrier pair

$$\eta_{\mathrm{ext}} = -6.729586682888832, \qquad \sigma_{\mathrm{ext}} = 8.154061112725994,$$

with ordered gap values

$$\gamma_{21} = 5.33160859254774, \qquad \gamma_{32} = 2.822452520178255, \qquad \kappa_{\text{ext}} = -4.59605680397234,$$

and centered logs

$$E_{\text{log,centered}} = [-4.495223235091244,\ 0.836385357456495,\ 3.6588378776347503].$$

Against the charged references, this continuation-centered shape has residual norm

$$\left\| E^{\text{cont}}_{\text{log,centered}} - E^{\text{ref}}_{\text{log,centered}} \right\| \approx 2.13 \times 10^{-5},$$

so this compare-only branch is a near-exact centered-shape closure up to one common absolute scale. It does not promote the theorem lane: the centered charged shape is nearly exact on this branch, but the public affine absolute anchor is external. The compare-only common scale required for exact absolute masses is

$$g_e^\star = 0.04577885783568762.$$

Equivalently, relative to the stored shared-budget seed

$$g_{\text{ch}}^{\text{shared}} = 0.9231656602589082,$$

the missing affine absolute anchor on the determinant-line route would have to take the compare-only value

$$\Delta_e^{\text{abs},\star} = \log \frac{g_{\text{ch}}^{\text{shared}}}{g_e^\star} = 3.003986333402356.$$

This pins down the remaining public closure burden sharply: the compare-only branch nearly solves the centered charged shape, but the theorem-grade derivation lacks promotion of the latent candidate $\widehat{C}_e^{\text{cand}}$ to theorem-grade $\widehat{C}_e$, and then one affine-covariant absolute anchor A_{ch} that would turn that centered readback into public charged masses on the theorem lane.

11 Neutrino Family Derivation

The neutrino derivation is the strongest positive flavor-side closure after the Higgs/top critical stage. The intrinsic isotropic branch is excluded exactly. On the declared weighted-cycle theorem branch, the PMNS/hierarchy shape lands in the observed regime, the bridge-rigidity theorem emits C_ν, absolute attachment fixes the paper-facing amplitude B_ν and λ_ν, and that branch emits the absolute eV-scale family. The older one- and two-parameter adapters remain compare-only diagnostic continuations beneath that theorem branch.

On that weighted-cycle theorem branch the paper emits

$$\theta_{12} = 34.225904631810025^\circ, \qquad \theta_{23} = 49.72282845058266^\circ, \qquad \theta_{13} = 8.686355527700156^\circ,$$

$$\delta_{\text{PMNS}} = 305.58061231449796^\circ, \qquad J = -0.02753115613565372,$$

and the dimensionless hierarchy invariant

$$\frac{\Delta m_{21}^2}{\Delta m_{32}^2} = 0.030721110097966534.$$

The exponent law is fixed by the positive transport-load segment between $\chi = 1+\epsilon$ and $1+\gamma_{1/2}$: on a one-dimensional affine segment, the balanced selector and the least-distortion selector for any positive translation-invariant quadratic form coincide at the midpoint, so

$$D_\nu = \frac{\chi + (1+\gamma_{1/2})}{2}, \qquad p = 1+\gamma+\frac{\epsilon}{D_\nu}.$$

On the emitted positive selector segment there is a stronger compare-only two-parameter adapter: solving τ_ν against the representative central ratio and then solving λ_ν against Δm^2_{32} gives

$$(m_1, m_2, m_3) = (0.01745663295,\ 0.01948419960,\ 0.05308139066)\ \text{eV},$$

$$\begin{aligned}\Delta m^2_{21} &= 7.49\times 10^{-5}\ \text{eV}^2,\\ \Delta m^2_{31} &= 2.5129\times 10^{-3}\ \text{eV}^2,\\ \Delta m^2_{32} &= 2.438\times 10^{-3}\ \text{eV}^2.\end{aligned}$$

These exact central numbers remain compare-only because the proof-facing theorem lane is the weighted-cycle bridge-rigid branch with

$$C_\nu = 0.9994295999075177, \qquad P_\nu = 6.699825740519345, \qquad B_\nu = P_\nu C_\nu = 6.696004159297337,$$

and therefore

$$\lambda_\nu = \frac{m_{\star,\text{eV}}}{q_{\text{mean}}^{p_\nu}} P_\nu C_\nu = 1.7237014208357415,$$

$$m_i = \lambda_\nu \widehat{m}_i, \qquad \Delta m^2_{ij} = \lambda_\nu^2 \widehat{\Delta}_{ij}.$$

The normalized same-label overlap-defect weight section $qbar_e$ is closed beneath this step, and on the isotropic branch the same-label phase-cocycle theorem, the selector-centered common-refinement bundle descent, and the finite-angle centered edge-norm theorem are also closed, so the exact two-parameter positive-segment adapter, the bridge corridor, and the bridge-coordinate sidecars are diagnostic surfaces beneath the bridge-rigidity invariant C_ν and its induced paper-facing amplitude B_ν rather than proof-facing ingredients.

$$C_\nu := \frac{B_\nu}{I_\nu^{1/2}\,\widehat{\text{ratio}}^{\,1/2}\,\text{sum_defect}^{-1}}$$

above the emitted proxy and closed normalizer. The attached stack also factors exactly through $q_e = q_{\text{mean}} qbar_e$, so a $qbar_e$-only collapse law for the bridge factor does not follow from the theorem surface stated here. The best closed constructive local object beneath that bridge is the defect-weighted same-label edge family $q_e = \sqrt{g_e d_e}$ together with the induced μ_e family, but that family sits below C_ν and the induced paper-facing amplitude B_ν rather than replacing them. The weighted-cycle branch fixes the PMNS data, the hierarchy ratio, and the scale-free mass normal form; on that branch the bridge-rigidity theorem fixes C_ν above the emitted proxy P_ν, and absolute attachment then emits the absolute eV-scale family.

On the same-label neutrino-only branch, the centered eta-class is exactly S_3-isotropic, so the same-label data are edge-constant and the solar $1-2$ split cannot open there by itself. The first solar mover therefore has to come from realized flavor-side same-label gap/defect readback, not from another neutrino-only selector tweak.

11.1 Intrinsic neutrino eta-chain

The proof-facing input is smaller than the raw same-label matrix payload. It compresses to the same-label scalar certificate

$$(g_e,\ \omega_e)_{e\in\{12,23,31\}}, \qquad \omega_e = \text{same-label overlap}_e^2, \qquad d_e = 1-\omega_e,$$

modulo one common scale. From that certificate one forms

$$q_e = \sqrt{g_e d_e}, \qquad \eta_e = \log q_e - \frac{1}{3}\sum_f \log q_f, \qquad e\in\{12,23,31\},$$

the intrinsic selector depends only on the centered class

$$[\eta_e] \in \mathbb{R}^3/\langle(1,1,1)\rangle.$$

Equivalently one may work with any positive normalized family

$$\mu_e = \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

Common scaling cancels identically, so the intrinsic selector is determined by the centered eta-class alone.

On the isotropic neutrino-only branch one has

$$(\eta_{12},\eta_{23},\eta_{31}) = (0,0,0),$$

which emits the intrinsic masses

$$(m_1,m_2,m_3) = (2.3986448447627196,\ 2.3986448447627196,\ 2.590074050773907)\times 10^{-12}\ \text{GeV},$$

with

$$\Delta m_{21}^2 = 0, \qquad \Delta m_{31}^2 = 9.549864971855843\times 10^{-25}\ \text{GeV}^2.$$

This is the exact neutrino-only isotropy obstruction behind the remaining solar-splitting gap.

Theorem 11.1 (Same-label scalar certificate suffices for intrinsic neutrino masses)**.** *Assume the same-label gap and overlap scalars are emitted on all realized arrows. Then the full intrinsic neutrino mass-eigenstate bundle factors through the scalar certificate*

$$(g_e,\omega_e)_{e\in\{12,23,31\}}$$

or equivalently through the centered eta-class $[\eta_e]$. No additional raw matrix payload is needed to form the intrinsic selector, the depressed-cubic spectrum, the ordered splittings, or the intrinsic neutrino-side basis.

Theorem 11.2 (Exact principal-branch selector from the centered eta-class)**.** *Fix the cycle sum*

$$\Omega = \psi_{12}+\psi_{23}+\psi_{31}, \qquad |\Omega| < \frac{3\pi}{2},$$

and a positive weight family μ_e. On the principal branch $\psi_e \in (-\pi/2,\pi/2)$, the affine energy

$$A(\psi) = \sum_e \mu_e(1-\cos\psi_e)$$

has a unique minimizer. It is given by

$$\psi_e^\star = \arcsin(\lambda/\mu_e),$$

where λ is the unique solution of

$$\sum_e \arcsin(\lambda/\mu_e) = \Omega, \qquad \lambda \in (-\min_e \mu_e, \min_e \mu_e).$$

Proof. The Euler–Lagrange equations are $\mu_e \sin\psi_e = \lambda$. On the principal branch the scalar function

$$F(\lambda) = \sum_e \arcsin(\lambda/\mu_e)$$

is strictly increasing because

$$F'(\lambda) = \sum_e \frac{1}{\sqrt{\mu_e^2 - \lambda^2}} > 0.$$

Its range is $(-3\pi/2, 3\pi/2)$, so $F(\lambda) = \Omega$ has a unique solution. Strict convexity follows because the Hessian of A is

$$\nabla^2 A = \mathrm{diag}(\mu_e \cos\psi_e),$$

which is positive definite on the principal branch and remains positive definite after restriction to the affine plane $\sum_e \psi_e = \Omega$. □

Let

$$a = m_\star = \frac{v^2}{\mu_u}, \qquad \rho = |(M_0)_{12}|.$$

Then the intrinsic Majorana matrix is

$$M(\eta) = \begin{pmatrix} a & \rho e^{i\psi_{12}} & \rho e^{i\psi_{31}} \\ \rho e^{i\psi_{12}} & a & \rho e^{i\psi_{23}} \\ \rho e^{i\psi_{31}} & \rho e^{i\psi_{23}} & a \end{pmatrix}.$$

Theorem 11.3 (Exact intrinsic spectral cubic)**.** *Let $H = M^\dagger M$. Then*

$$H = dI + T, \qquad d = a^2 + 2\rho^2,$$

where T has zero diagonal and off-diagonals

$$\begin{aligned} x_{12} &= 2a\rho\cos\psi_{12} + \rho^2 e^{i(\psi_{23}-\psi_{31})}, \\ x_{23} &= 2a\rho\cos\psi_{23} + \rho^2 e^{i(\psi_{31}-\psi_{12})}, \\ x_{13} &= 2a\rho\cos\psi_{31} + \rho^2 e^{i(\psi_{12}-\psi_{23})}. \end{aligned}$$

Define

$$P = |x_{12}|^2 + |x_{23}|^2 + |x_{13}|^2, \qquad Q = \Re(x_{12}x_{23}\overline{x_{13}}).$$

Then the eigenvalues λ_k of T are exactly the three real roots of

$$\lambda^3 - P\lambda - 2Q = 0,$$

and the intrinsic squared masses are

$$m_k^2 = d + \lambda_k.$$

Equivalently,

$$\lambda_k = 2\sqrt{\frac{P}{3}}\cos\left(\frac{1}{3}\arccos\left(\frac{3\sqrt{3}\,Q}{P^{3/2}}\right) - \frac{2\pi k}{3}\right).$$

Corollary 11.4 (Intrinsic eta-chain closure)**.** *Once the centered eta-class is emitted at the flavor boundary, the intrinsic neutrino branch emits*

$$\nu_1, \nu_2, \nu_3, \qquad \Delta m_{21}^2, \Delta m_{31}^2, \Delta m_{32}^2,$$

the ordering, and the intrinsic neutrino-side basis U_ν with no PMNS import and no flavor-label leakage.

11.2 Perturbative laws around the isotropic point

Write

$$\psi_e = \varphi + \delta_e, \qquad \delta_{12} + \delta_{23} + \delta_{31} = 0.$$

At first order in the centered eta-class,

$$\delta_e = -\tan\varphi\, \eta_e + O(\eta^2).$$

Define

$$\sigma^2 = \frac{2}{3}\sum_e \delta_e^2.$$

Then

$$m_{1,2}^2 = m_d^2 \mp 2a\rho \sin\varphi\, \sigma + O(\delta^2),$$

so

$$\Delta m_{21}^2 = 4a\rho \sin\varphi\, \sigma + O(\delta^2) = 4a\rho \frac{\sin^2\varphi}{\cos\varphi}\sqrt{\frac{2}{3}\sum_e \eta_e^2} + O(\eta^2).$$

The ordered atmospheric gaps obey the corrected first-order laws

$$\Delta m_{31}^2 = \Delta_{\text{atm,iso}} + \frac{1}{2}\Delta m_{21}^2 + O(\delta^2), \qquad \Delta m_{32}^2 = \Delta_{\text{atm,iso}} - \frac{1}{2}\Delta m_{21}^2 + O(\delta^2),$$

so the first-order invariant atmospheric object is the collective-to-doublet centroid gap

$$\Delta_{\text{cent}} = m_3^2 - \frac{1}{2}(m_1^2 + m_2^2) = \Delta_{\text{atm,iso}} + O(\delta^2).$$

Its quadratic shift is

$$\delta\Delta_{\text{cent}} = -a\rho\,\sigma^2 \frac{a(4\cos^2\varphi - 1) + 6\rho\cos\varphi}{2a\cos\varphi + \rho} + O(\delta^3).$$

The largest-mass singular vector obeys the first-order deformation law

$$u_3 = u + \kappa \begin{pmatrix} \delta_{23} \\ \delta_{31} \\ \delta_{12} \end{pmatrix} + O(\delta^2), \qquad u = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad \kappa = \frac{\sqrt{3}\,(2a\sin\varphi + 3i\rho)}{9(2a\cos\varphi + \rho)}.$$

Equivalently,

$$u_3 = u - \tan\varphi\, \kappa \begin{pmatrix} \eta_{23} \\ \eta_{31} \\ \eta_{12} \end{pmatrix} + O(\eta^2).$$

One exact demonstration uses centered eta-class

$$(\eta_{12}, \eta_{23}, \eta_{31}) = (0.13631072512014578,\ -0.18362987264261327,\ 0.0473191475224675),$$

and emits intrinsic masses

$$(m_1, m_2, m_3) = (2.3929601069646055,\ 2.4048200109774875,\ 2.589606227283229) \times 10^{-12}\ \text{GeV},$$

with

$$\Delta m_{21}^2 = 5.690121167370743 \times 10^{-26}\ \text{GeV}^2, \qquad \Delta m_{31}^2 = 9.798023388600230 \times 10^{-25}\ \text{GeV}^2.$$

These are exact outputs of the intrinsic eta-chain once the centered eta-class is supplied. They are not yet flavor-labeled OPH rows.

11.3 Weighted-cycle bridge rigidity and absolute attachment

The weighted-cycle route closes the PMNS and hierarchy steps. On that lane the bridge-rigidity theorem fixes the reduced invariant C_ν above the emitted proxy P_ν, with $B_\nu = P_\nu C_\nu$ retained only as the paper-facing amplitude parameterization:

$$\begin{aligned} C_\nu &= G^2 Q S^{-1/2} = 0.9994295999075177, \\ P_\nu &= 6.699825740519345, \\ B_\nu &= P_\nu C_\nu = 6.696004159297337. \end{aligned}$$

The absolute-attachment theorem therefore emits

$$\begin{aligned} \lambda_\nu &= \frac{m_{\star,\text{eV}}}{q_{\text{mean}}^{p_\nu}} P_\nu C_\nu = 1.7237014208357415, \\ m_i &= \lambda_\nu \hat{m}_i, \\ \Delta m_{ij}^2 &= \lambda_\nu^2 \widehat{\Delta m_{ij}^2}. \end{aligned}$$

No hidden discrete branch remains on that lane. The one-parameter atmospheric-anchor slice and the stronger two-parameter positive-segment adapter remain only as diagnostic compare-only continuations beneath the bridge-rigidity invariant C_ν and its induced paper-facing amplitude B_ν.

12 Hadrons, QCD, and the Emergence of Ordinary Matter

The hadron derivation is the most operationally demanding part of the whole particle story. The frontier is one production backend export bundle with publication-complete manifest provenance and real correlator arrays, followed by executed unquenched production data plus production statistical and systematic control. The declared particle roadmap does not include hadron-mass closure, because that nonperturbative execution burden is not a realistic target within the current project compute scope. Accordingly, the hadron rows are not derived outputs of this paper. The derivation nevertheless has a complete mathematical execution bridge, a deterministic runtime receipt, a full writeback/evaluation path, and a surrogate end-to-end executable bundle that closes the declared execution surface without pretending to be production lattice QCD.

12.1 Seeded $2+1$ family and unquenched measure

Let

$$m_l := \frac{m_u + m_d}{2}, \qquad \rho_l := \frac{m_u + m_d}{2\,\Lambda^{(3)}_{\overline{\mathrm{MS}}}}, \qquad \rho_s := \frac{m_s}{\Lambda^{(3)}_{\overline{\mathrm{MS}}}}.$$

The seeded fixed-physics family is

$$a\Lambda_n = a\Lambda_{\mathrm{seed}}\, 2^{-n}, \qquad a\Lambda_{\mathrm{seed}} = \sqrt{\rho_l \rho_s},$$

$$am_l^{(n)} = a\Lambda_n \rho_l, \qquad am_s^{(n)} = a\Lambda_n \rho_s,$$

$$\beta_n = 6 + \frac{9}{2\pi^2} \log \frac{1}{a\Lambda_n}, \qquad L_n = \left\lceil \frac{\lambda_L^{\mathrm{target}}}{a\Lambda_n} \right\rceil, \qquad T_n = \left\lceil \frac{\lambda_T^{\mathrm{target}}}{a\Lambda_n} \right\rceil.$$

The unquenched ensemble measure is

$$d\mu_n(U) = Z_n^{-1} \exp[-S_g(U;\beta_n)] \det D_l(U; am_l^{(n)})^2 \det D_s(U; am_s^{(n)})\, dU.$$

On this branch, $N_f = 2+1$ and QED is off.

12.2 Runtime receipt and deterministic cfg/source contract

The emitted execution law is

$$U_{n,c} = K_n^{N_{\mathrm{therm}} + (\mathrm{cfg_index})\, N_{\mathrm{sep}}}(U_{\mathrm{cold}}; \mathrm{seed}_{n,c}),$$

with stop-time formula

$$t_{\mathrm{stop}}(n,c) = N_{\mathrm{therm}} + (\mathrm{cfg_index})\, N_{\mathrm{sep}}.$$

The deterministic cfg seed law is

$$\mathrm{seed}_{n,c} = \mathrm{bytes.fromhex}(\mathrm{cfg_seed_hash}_{n,c}),$$

with

$$\mathrm{cfg_seed_hash} = \mathrm{SHA256}\Big(\mathrm{Serialize}\big(\mathrm{ensemble_id}, \beta, L, T, am_l, am_s, \mathrm{cfg_index}\big)\Big).$$

The source set is fixed as

$$S_{n,c} = \{[0,0,0,0],\ [L_n//2, L_n//2, L_n//2, T_n//2]\}.$$

The non-null external runtime receipt used in the surrogate execution is

$$N_{\mathrm{therm}} = 2048, \qquad N_{\mathrm{sep}} = 512.$$

These values are surrogate execution inputs only; they are not claimed as theorem outputs.

The production geometry summary makes the remaining runtime boundary concrete. On the emitted seeded $2+1$ family there are three ensembles and six configurations total. If one stores all four links at every site as full double-complex 3×3 matrices, the naive raw gauge storage estimate is

$$576 \text{ bytes/site},$$

which gives total naive raw gauge storage

$$2.80071464105088 \times 10^{14} \text{ bytes}$$

over the production schedule. The normalized backend correlator dump needed by the analysis layer is tiny:

$$195264 \text{ bytes}$$

for the full π_{iso}, $N_{\text{iso,dir}}$, and $N_{\text{iso,ex}}$ payload on the production schedule. The remaining hadron requirement is the backend export bundle that would feed the normalized dump from real production execution.

12.3 Operational reading of the remaining computation

It is useful to separate the lightweight analysis layer from the expensive numerical layer. Informally, the repository-side control plane is explicit: the seeded ensemble family, the deterministic cfg/source contract, the export manifest, the jackknife evaluator, the forward-window selector, and the publication-budget schema are all explicit. The missing step is therefore not another bookkeeping script but the physical generation of dynamical gauge configurations and measured correlator arrays on the fixed $N_f = 2 + 1$, QED-off branch.

Technically, the remaining production computation is the standard lattice-QCD stable-channel workflow on the emitted family: run unquenched RHMC/HMC for the three seeded ensembles; for each realized cfg and fixed source solve the clover-improved Wilson light/strange systems; construct the zero-momentum π_{iso}, $N_{\text{iso,dir}}$, and $N_{\text{iso,ex}}$ two-point sequences; write the backend bundle; convert that bundle into the repo-side production dump; then let the existing jackknife and forward-window machinery emit $am_{X,\text{ground}}$, R_X, $m_X[\text{GeV}]$, and the published σ_{stat}, δ_{cont}, δ_{vol}, and δ_χ fields. In that sense the hadron lane is analysis-complete but execution-incomplete.

The engineering burden is therefore asymmetric. Local execution is sufficient for manifest generation, surrogate validation, and downstream readout tests, because those stages operate on a tiny correlator payload. The physical branch is different: the dominant cost is upstream gauge-field generation and Dirac solves, not the final analysis dump. This is therefore a classical high-performance-computing workload, not a special quantum-computing shortcut. Quantum-cloud quantum-cloud hardware can support toy lattice-gauge demonstrations, but it is not a realistic execution target for full $3 + 1$-dimensional dynamical $SU(3)$ lattice-QCD hadron spectroscopy. So the direct deployment target is a multi-node HPC cluster or cloud HPC allocation instead of a laptop, ordinary workstation, or quantum-cloud device. As an operational estimate, a backend adapter and first real stable-channel pilot dump are a week-scale software task, whereas publishable $\pi_{\text{iso}}/N_{\text{iso}}$ systematics are month-scale numerical work and depend strongly on the available allocation, solver quality, and whether one stays on the stable-channel branch or also opens the later ρ-scattering and isospin-breaking program. For that reason, the declared particle roadmap stops at the hadron execution boundary rather than treating production hadron masses as planned deliverables.

12.4 Stable-channel correlators, effective masses, and forward windows

The pion stable channel is

$$p_\pi^{(n,c,s)}(t) = \sum_x \Re \operatorname{tr}_{c,\text{spin}} \left[\gamma_5 S_l(x;s)\gamma_5 S_l(s;x)\right].$$

The nucleon stable channel is

$$p_{N,\text{dir}}^{(n,c,s)}(t) = \sum_x G_d, \qquad p_{N,\text{ex}}^{(n,c,s)}(t) = \sum_x G_x,$$

$$p_N^{(n,c,s)}(t) = p_{N,\mathrm{dir}}^{(n,c,s)}(t) - p_{N,\mathrm{ex}}^{(n,c,s)}(t).$$

Cfg/source and ensemble averaging are

$$\bar{p}_X^{(n,c)}(t) = \frac{1}{|S_{n,c}|} \sum_{s \in S_{n,c}} p_X^{(n,c,s)}(t),$$

$$C_X^{(n)}(t) = \frac{1}{|C_n|} \sum_{c \in C_n} \bar{p}_X^{(n,c)}(t).$$

The effective-mass laws are

$$am_{\mathrm{eff},\pi}(t) = \log \frac{C_\pi(t)}{C_\pi(t+1)}, \qquad am_{\mathrm{eff},N}(t) = \log \frac{|C_N(t)|}{|C_N(t+1)|}.$$

The forward window is

$$W_n = \{t : 1 \le t+1 < \lfloor T_n/2 \rfloor\}.$$

The evaluator monitors log-convexity,

$$R_{\log\mathrm{conv},\pi}(t) = C_\pi(t)^2 - C_\pi(t-1)C_\pi(t+1),$$

$$R_{\log\mathrm{conv},N}(t) = |C_N(t)|^2 - |C_N(t-1)|\,|C_N(t+1)|,$$

tail-drop,

$$D_X(t) = am_{\mathrm{eff},X}(t) - am_{\mathrm{eff},X}(t+1),$$

and mirror suppression,

$$M_X(t) = \exp[-am_{\mathrm{eff},X}(t)(T_n - 2t)].$$

The selected forward window is the longest contiguous run satisfying finite effective masses, nonnegative log-convexity up to tolerance, nonnegative tail-drop up to tolerance, mirror suppression below threshold, and local plateau flatness. The candidate ground-state mass is then the weighted window average

$$am_{X,\mathrm{ground}} = \frac{\sum_{t \in W_X^{\mathrm{sel}}} w_t\, am_{\mathrm{eff},X}(t)}{\sum_{t \in W_X^{\mathrm{sel}}} w_t}, \qquad w_t = \frac{1}{\max(\sigma_t^2, \varepsilon)}.$$

12.5 Statistics, systematics, and dimensional readout

Delete-1 jackknife is performed over the cfg axis after source averaging inside each cfg. With n_{cfg} configurations and integrated autocorrelation time $\tau_{\mathrm{int,cfg}}$, the effective cfg count is

$$n_{\mathrm{eff,cfg}} = \frac{n_{\mathrm{cfg}}}{2\,\tau_{\mathrm{int,cfg}}}.$$

The published statistical error is

$$\sigma_{\mathrm{stat},X} = \mathrm{JKstderr}(am_{X,\mathrm{ground}}).$$

The machine-readable systematics field uses

$$\sigma_{\mathrm{sys},X} = \sqrt{\delta_{\mathrm{cont},X}^2 + \delta_{\mathrm{vol},X}^2 + \delta_{\chi,X}^2}.$$

Continuum, volume, and chiral proxies are encoded as

$$R_X^{(n)} = \frac{am_X^{(n)}}{a\Lambda_n}, \qquad R_X^{(n)} \approx R_X(0) + c_X(a\Lambda_n)^2,$$

$$\delta_{\mathrm{cont},X}^{(n)} = a\Lambda_n \left| R_X^{(n)} - R_X(0) \right|,$$

$$\delta_{\mathrm{vol},\pi}^{(n)} = am_\pi^{(n)} \frac{e^{-am_\pi^{(n)} L_n}}{\max(am_\pi^{(n)} L_n, 1)},$$

$$\delta_{\mathrm{vol},N}^{(n)} = am_N^{(n)} \frac{e^{-am_\pi^{(n)} L_n}}{\max(am_\pi^{(n)} L_n, 1)},$$

$$\delta_{\chi,\pi}^{(n)} = a\Lambda_n \left| R_\pi^{(n)} - \langle R_\pi \rangle_n \right|,$$

$$Q_N^{(n)} = \frac{R_N^{(n)}}{R_\pi^{(n)}}, \qquad \delta_{\chi,N}^{(n)} = a\Lambda_n \langle R_\pi \rangle_n \left| Q_N^{(n)} - \langle Q_N \rangle_n \right|.$$

These are surrogate publication proxies, not production physical systematics. Given a ground-state candidate,

$$R_X = \frac{am_{X,\mathrm{ground}}}{a\Lambda_{\overline{\mathrm{MS}}}^{(3)}}, \qquad m_X[\mathrm{GeV}] = R_X\, \Lambda_{\overline{\mathrm{MS}}}^{(3)}[\mathrm{GeV}],$$

with

$$\Lambda_{\overline{\mathrm{MS}}}^{(3)} = 0.3344017073 \text{ GeV}.$$

12.6 Surrogate execution bundle and frontier

The surrogate execution evolves a latent state $z \in \mathbb{R}^d$ with Hamiltonian

$$H(z,p) = S(z) + \frac{1}{2}\sum_i p_i^2,$$

where

$$S(z) = \frac{1}{2}\sum_i \omega_i^2 z_i^2 + \lambda_4 \sum_i z_i^4 + \kappa \sum_i (z_{i+1} - z_i)^2 + 2\alpha_l \sum_i \log(\mu_l^2 + z_i^2) + \alpha_s \sum_i \log(\mu_s^2 + \tfrac{1}{2} z_i^2).$$

Leapfrog integration plus Metropolis accept/reject gives the surrogate HMC update

$$(z,p) \mapsto (z',p'), \qquad P_{\mathrm{acc}} = \min(1, e^{-\Delta H}).$$

This kernel is not physical lattice-QCD RHMC/HMC. It is a deterministic executable surrogate honoring the emitted receipt and seed law.

For validation of the execution bridge, the surrogate locks the ground-state masses to the hadron audit proxy values

$$m_{\pi,\mathrm{proxy}} = 0.13497682776768472 \text{ GeV}, \qquad m_{N,\mathrm{iso,proxy}} = \frac{m_p + m_n}{2} = 0.93891875434 \text{ GeV}.$$

Thus

$$am_{\pi,\mathrm{proxy}}^{(n)} = \frac{m_{\pi,\mathrm{proxy}}}{\Lambda_{\overline{\mathrm{MS}}}^{(3)}[\mathrm{GeV}]}\, a\Lambda_n,$$

$$am^{(n)}_{N,\mathrm{iso,proxy}} = \frac{m_{N,\mathrm{iso,proxy}}}{\Lambda^{(3)}_{\overline{\mathrm{MS}}}[\mathrm{GeV}]}\, a\Lambda_n.$$

The surrogate correlator is

$$C^{\mathrm{sur}}_X(t) = A_{X,0}e^{-am_X t} + A_{X,1}e^{-am_{X,\mathrm{ex}}t} + A_{X,\mathrm{mir}}e^{-am_X(T_n-t)}$$

up to a multiplicative correlated noise factor. Direct and exchange nucleon pieces are written as

$$C^{\mathrm{sur}}_{N,\mathrm{dir}}(t) = f_{\mathrm{dir}}\, C^{\mathrm{sur}}_N(t), \qquad C^{\mathrm{sur}}_{N,\mathrm{ex}}(t) = (f_{\mathrm{dir}} - 1)\, C^{\mathrm{sur}}_N(t),$$

so that

$$C^{\mathrm{sur}}_N(t) = C^{\mathrm{sur}}_{N,\mathrm{dir}}(t) - C^{\mathrm{sur}}_{N,\mathrm{ex}}(t).$$

On the finest surrogate ensemble, the resulting diagnostic candidates are

$$m^{\mathrm{sur}}_{\pi,\mathrm{iso}} = 0.135039383836\ \mathrm{GeV}, \qquad m^{\mathrm{sur}}_{N,\mathrm{iso}} = 0.938960210578\ \mathrm{GeV},$$

with worst absolute error approximately 6.26×10^{-5} GeV across the surrogate stable-channel family. These numbers validate the emitted execution bridge. They are not promotable production hadron predictions.

So the hadron derivation closes the full

$$\text{receipt} \to \text{execution} \to \text{writeback} \to \text{evaluation} \to \text{budgets} \to \text{forward-window summary}$$

path on executed surrogate data, while the remaining physical closure requires production unquenched RHMC/HMC, real Dirac solves and baryon contractions, production autocorrelation studies, and production continuum / finite-volume / chiral systematics.

13 Observer-Centric Particle Ontology and Measurement

This paper cannot stop at masses and couplings. OPH is observer-centric at the level of its basic ontology, so a particle chapter also has to explain what a particle *is* in this language and how a measurement turns an excitation surface into an actual observed state. Fortunately, this is one of the places where the OPH theorem surface contains real mathematical content rather than only interpretation.

13.1 Observer patches, overlap consistency, and particle data

Ref. [2] formulates the basic kinematic picture in patch-net language: each observer patch carries a local state, neighboring patches compare a shared interface alphabet, and a global state is physically admissible exactly when neighboring projections agree on the overlap. The algebraic OPH version of the same statement is Axioms 5.1 and 5.2: local physical data are carried by patch algebras $\mathcal{A}(P)$, and those local states must agree on shared subalgebras whenever patches overlap.

This matters for particles because OPH does not begin with a single absolute global particle basis. It begins with patch-local algebras, overlap-visible observables, edge sectors, and transport data. A particle row in the reported derivation is therefore not best understood as an isolated primitive label. It is a readout claim about a stable or candidate-stable excitation structure visible to a family of observer patches and consistent on their overlaps.

In the observer formulation, an observer is represented by

$$O = (P, \mathcal{A}(P), \rho, R),$$

where P is the observer patch, $\mathcal{A}(P)$ its local algebra, ρ the local state, and R the record algebra. The key point for the particle derivation is that R is not an extra classical layer stapled onto quantum theory. On the exact fixed-cutoff measurement surface it is generated by central record projectors, and practical readout may use approximately commuting projectors that are close to that central reference algebra. That is what makes the observer-facing record surface shareable across overlapping observer descriptions without violating the usual no-cloning constraints on generic quantum states.

13.2 Record algebras and definite outcomes

The integrated measurement appendices and Ref. [3] make the measurement interface precise at fixed cutoff. On the declared operational surface, the completed write/verify slice carries a finite commutative central record algebra. Practical readout may instead use projectors Q_a on the same declared slots with commuting central reference projectors $\widehat{Q}_a$, where

$$\delta_{\mathrm{rec}} := \max_a \|Q_a - \widehat{Q}_a\|.$$

Then

$$\|[Q_a, Q_b]\| \le 4\,\delta_{\mathrm{rec}},$$

and, whenever $\|\widetilde{\rho} - \rho\|_1 \le \varepsilon$,

$$\Big|\mathrm{Tr}(\widetilde{\rho} Q_a) - \mathrm{Tr}(\rho \widehat{Q}_a)\Big| \le \varepsilon + \delta_{\mathrm{rec}}.$$

On the explicit fixed-cutoff screen architecture, the exact record algebra after a completed write/verify cycle is

$$\mathcal{Z}_{\mathrm{rec}}(t) = \mathrm{Alg}\Big(\{P_{m_I^{(J)}(t)}\}_{I\neq J}, \{P_{r_I^{\mathrm{bulk}}(t)}\}_I, \{\Pi_\alpha^{(IJ)}(t)\}_{I<J,\alpha}\Big).$$

Ref. [3] proves that, on the declared operational measurement surface, $\mathcal{Z}_{\mathrm{rec}}(t)$ is commutative and central for the readout instrument being used.

This fixed-cutoff centrality result is what supports definite outcomes without adding a second ontology. The integrated measurement ledger phrases the same idea through edge-center decomposition: on a fixed collar with exact Markov structure, the state splits into superselection blocks,

$$\rho_{ABC} = \bigoplus_j q_j\, \rho^{(j)}_{Ab_L^{(j)}} \otimes \rho^{(j)}_{b_R^{(j)}C},$$

and the label j is classical center data. The supplement then goes one step further: in the physical algebra there are no interference observables between different sector blocks. So once a particle detection event has been recorded in the observer-accessible center/record algebra, the accessible physics is organized as a classical mixture over those blocks rather than as a superposed measurement record.

Accordingly, a measured particle family or excitation channel should be thought of as an observer-accessible sector or record value carried by the central measurement surface, not as a mysterious absolute collapse of the full universe-state into a preferred basis chosen by hand.

13.3 Born probabilities and post-measurement states

The fixed-cutoff measurement package is also explicit about probabilities and updates. For every event E in the sigma algebra generated by $\mathcal{Z}_{\mathrm{rec}}(t)$, the measurement probability is

$$\mathbb{P}_t(E) = \mathrm{Tr}\big(\rho_t P_E\big),$$

where $P_E \in \mathcal{Z}_{\mathrm{rec}}(t)$ is the projector for that event. Conditioning on the event then produces the post-measurement state

$$\rho_t|_E = \frac{P_E \rho_t P_E}{\mathrm{Tr}(\rho_t P_E)}.$$

In the supplement, the same statement appears as the Lüders update on the record algebra. In the synthesis paper *Observers Are All You Need* [1], this fixed-cutoff Born/Lüders package is imported as theorem-bearing rather than interpretive rhetoric.

This directly answers the particle-state question in the present framework. Measurements give particles actual states by conditioning the observer-accessible state on the central record algebra. A detector click, a stable overlap-sector readout, or a pointer value does not merely announce a pre-existing classical label; it defines the conditioned state for subsequent observer-accessible physics. If the same event is re-read without any accepted repair move touching its support, the result in Ref. [3] shows that the re-read probability is 1. So the measurement interface is well defined and operationally auditable.

13.4 Particle identity as transport-stable structure

Once one asks "which event happened?" and then "which particle family is this?", the flavor derivation becomes essential. The active flavor derivation does not start with the names electron, muon, tau lepton, or up quark built into the fundamental observable. It starts with transport kernels, generation-bundle data, same-label eigenline transport, overlap-edge cocycles, and a persistent flavor observable carrying intrinsic labels $f1, f2, f3$. This is an important discipline condition in the paper. At the deepest flavor surface used here, family identity is transport-stable intrinsic structure first and named experimental family assignment second.

That is why the shared Yukawa/excitation dictionary matters so much even though it is not itself the leading missing object. The flavor pipeline exports a substantial invariant base: projectors, spectral gaps, pair suppressions, cycle phases, and common sector-response objects. What remains unfinished is the map from that intrinsic base to the named low-energy fermion families. The quark derivation emits named rows because the forward branch is far enough along to support direct numerical comparison there; the charged-lepton derivation does not do so at theorem grade, and the neutrino derivation does so only on its weighted-cycle theorem branch rather than on a fully flavor-labeled closure surface, even though they use the same downstream family machinery.

So in the strongest justified sense, particle identity in OPH is a refinement-stable observer-accessible transport pattern whose named interpretation is inherited from the available dictionary and comparison surfaces. Where that dictionary is still open, the paper should say so explicitly.

13.5 Scope boundary of the measurement chapter

This whole chapter has to preserve one final distinction. The fixed-cutoff measurement interface is theorem-bearing on the microphysics surface: central record algebra, Born probabilities, Lüders conditioning, and repeated-read stability are written there as fixed-cutoff statements. What is *not* yet proved at the same level is the stronger global observer-continuation or strange-loop story sometimes discussed in broader OPH notes. This paper should therefore use the fixed-cutoff measurement package directly, while leaving the larger metaphysical closure story out of the main technical chain.

14 Discussion: Matter, Antimatter, Supersymmetry, and Other Open Questions

This section is intentionally not written in the same voice as the structural and quantitative sections above. The OPH theorem surface distinguishes sharply between recovered-core results, calibration branches, secondary quantitative branches, and continuations. Matter versus antimatter, supersymmetry, and several adjacent questions lie on that boundary. They are too important to omit from a particle-spectrum paper, but they are not yet part of the finished prediction surface in the same way that the bosonic and quark rows are.

14.1 Matter and antimatter

The Standard Model structural branch fixes the chiral gauge architecture within which matter and antimatter are defined. Once a chiral fermion representation is present, its conjugate representation gives the corresponding antiparticle content. In that limited sense, OPH explains why matter and antimatter both exist: they are paired by the realized gauge and chiral structure of the low-energy branch.

What is *not* yet structurally closed is the cosmological asymmetry between them. The integrated continuation notes record a baryogenesis story tied to the realized $\mathbb{Z}_6$ quotient and the electroweak topological channel. In that continuation, the entropy deficit of the $\mathbb{Z}_6$ quotient gives a suppression factor

$$\varepsilon = e^{-\ln 6} = \frac{1}{6},$$

and combining this with the 12-fermion electroweak baryon-violating channel gives the order-of-magnitude estimate

$$\varepsilon^{12} = 6^{-12} \approx 4.6 \times 10^{-10},$$

which is numerically close to the observed baryon asymmetry scale.

But the observer-side status notes are explicit about the status of that estimate: the baryon asymmetry scale remains a Phase-III deferred continuation, and suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism. The present derivation also does not expose a baryogenesis output row. So this paper can report the existing OPH continuation idea, but it should not present a sharp matter/antimatter abundance prediction as if it were a closed theorem or an emitted output.

The missing ingredients are specific: a concrete out-of-equilibrium epoch, justified defect/sphaleron production dynamics, derived CP-odd source terms, transport and washout control, and a freeze-out computation of the final asymmetry. The same continuation also notes that CP-odd overlap data are generically available because central overlap phases are sent to their inverses under CP, but that observation is not yet a baryogenesis theorem.

14.2 Why the present branch does not require supersymmetry

The D10 appendix below contains the dedicated gauge-coupling calibration discussion unification. Its main point is not that supersymmetry has been mathematically ruled out in every possible extension of the program. Its point is narrower and more relevant to this paper: the present branch does not *need* a supersymmetric partner spectrum in order to explain why unification-like running might appear.

At one loop, the integrated D10 appendix records the familiar fact that Standard Model beta-function coefficients do not produce successful naive unification, whereas MSSM-like coefficients do.

The OPH continuation idea is then that edge-sector heat-kernel weights can shift the effective beta-function coefficients in an MSSM-like direction through geometric or entropic sector multiplicity rather than through an actual low-energy superpartner spectrum. In that reading, "unification-like behavior" and "a supersymmetric particle zoo" come apart.

For this paper, the correct conclusion is therefore conditional and modest:

1. the derivation reconstructs the realized Standard Model branch without having to introduce a supersymmetric partner for every known field;
2. the integrated D10 appendix contains a continuation-level argument that some unification-style running features could arise from edge-sector structure rather than MSSM particle content;
3. none of this is yet a universal theorem that supersymmetry is impossible.

So if the paper asks "why is there no supersymmetry in the derived spectrum?", the best technical answer is: because the realized branch stops at the Standard Model content, and the existing unification discussion is trying to reproduce the relevant running behavior geometrically, without extending the particle content to a full supersymmetric multiplet structure.

14.3 Three generations, flavor closure, and what remains open

The structural Standard Model branch fixes $N_g = 3$ on the realized MAR-admissible branch. But this paper should not blur this result together with the much harder flavor closure problem. Three generations are structurally recovered; the full flavor dictionary, excitation map, CKM closure, and charged-lepton closure remain partially open, while the neutrino lane is closed only on its declared weighted-cycle theorem branch rather than on a fuller flavor-labeled closure surface. This is why the flavor chapter and the later family chapters spend so much time on constructive object boundaries instead of only on final numbers.

14.4 Ordinary matter and the missing hadron closure

A second distinction belongs in discussion form rather than in the results sections: the difference between "elementary rows exist" and "ordinary matter is fully derived." Ordinary visible matter is dominated by hadrons, especially protons and neutrons. The present OPH derivation has a serious hadron pipeline, but it lacks promotable hadron rows on the default surface. The route toward ordinary matter is architecturally present, and the full observed matter spectrum has not been numerically derived end to end.

14.5 How to read these open questions

The reader should therefore interpret this whole discussion section with the same rule used by *Recovering Relativity and Standard Model Structure from Observer Overlap Consistency* [4]. A failure in one of these continuation stories would not by itself undo the recovered-core gauge and gravity chain, nor would it erase the reported bosonic and quark rows. Charged-lepton, fuller flavor-labeled neutrino, and hadron closure theorems sit beyond the closed structural core. Their truth or failure does not alter the recovered-core gauge and gravity chain, and it does not erase the reported bosonic and quark outputs.

15 Conclusion

The particle-spectrum derivation is nontrivial and much more concrete than a qualitative aspiration. It is not yet a uniformly closed derivation of the full particle zoo. The structural theorem surface fixes the realized Standard Model branch and the exact massless carrier sectors. The derivation then emits quantitative bosonic rows through electroweak calibration and the Higgs/top critical stage, closes the selected-class quark theorem on the selected public physical quark frame class f_P, and closes the neutrino lane on its declared weighted-cycle theorem branch. Charged leptons, flavor-labeled neutrinos, and hadrons have derivation chains beyond that surface, and the charged-lepton and hadron lanes terminate at named constructive objects rather than promotable masses.

That asymmetry belongs on the page because it is the mathematical state described in this paper. The paper puts the finished structural results, the emitted quantitative rows, and the named theorem boundaries onto one common page. The visible gates sit above the shared excitation dictionary, not inside it: theorem-level charged-lepton closure, global classification of public quark frame classes beyond the selected-class theorem on f_P, extension of the weighted-cycle neutrino theorem branch to a fuller flavor-labeled neutrino surface, and the nonperturbative hadron branch. The same accounting also sharpens the local unification story: the bosonic P-driven trunk is explicit, and the gravity-side bridge to G is reduced to a small named exact-release package. This paper should be read as a technical derivation document with exact structural outputs, exact selected-class quark closure, and explicit theorem boundaries on the unfinished sectors.

A D10 Heat-Kernel Calibration Supplement

This appendix rehouses the D10 compact-group heat-kernel items from the former technical supplement. The theorem-bearing leaf for the same-overlap thermal/Casimir law now lives on the screen-microphysics paper surface at fixed cutoff. What is recorded here is the later compact-group / Peter–Weyl lift and beta-shift bookkeeping used by the particle-side calibration branch. The public D10 status remains explicitly Phase II rather than recovered-core Phase I.

A.1 One-Loop Calibration Frame

At one loop, if couplings unify at (M_U, α_U),

$$\alpha_i^{-1}(M_Z) = \alpha_U^{-1} + \frac{b_i}{2\pi} \ln \frac{M_U}{M_Z}.$$

On the live OPH lane these are the branch values $(M_U(P), \alpha_U(P))$ emitted by the forward D10 solve; they are not inferred by inverse readback from measured low-energy couplings. Writing $A_i := \alpha_i^{-1}(M_Z)$ and $L := \ln(M_U/M_Z)$, one finds

$$L = \frac{2\pi}{b_1 - b_2}(A_1 - A_2),$$

and the corresponding consistency relation for the D10 lane is

$$A_3^{\text{pred}} = \frac{b_3 - b_2}{b_1 - b_2} A_1 + \frac{b_1 - b_3}{b_1 - b_2} A_2.$$

This appendix keeps that algebra on the page as a calibration relation, not as a theorem that OPH has derived a supersymmetric UV spectrum.

A.2 Benchmark coefficient comparison

For the standard one-loop benchmark coefficients,

Model	(b_1, b_2, b_3)	$\alpha_s(M_Z)^{\text{pred}}$
SM	$(41/10, -19/6, -7)$	≈ 0.071
MSSM	$(33/5, 1, -3)$	≈ 0.116
Observed		0.1179 ± 0.0010

At one loop, the printed MSSM-style coefficients reproduce the observed closure far better than the plain SM coefficients. On the OPH lane this remains a benchmark matched by the D10 calibration branch, not a theorem that OPH has already derived a supersymmetric UV spectrum.

A.3 Heat-Kernel Sector Law

Read this subsection as the compact-group merge boundary above the fixed-cutoff microphysics theorem. MaxEnt plus the bi-invariant compact-group package yield the familiar $d_R e^{-tC_2(R)}$ law once the Peter–Weyl lift is supplied; they are not a second, independent proof of the finite-cutoff same-overlap Casimir branch.

Theorem A.1 (Heat-kernel edge-sector weights)**.** *Under MaxEnt with bi-invariant constraints on a compact Lie group G, the sector probabilities take heat-kernel form:*

$$p_R(t) \propto d_R\, e^{-tC_2(R)},$$

where $d_R = \dim R$ and $C_2(R)$ is the quadratic Casimir.

Lemma A.2 (Bi-invariant operators)**.** *If $G = \prod_i G_i$ is a compact semisimple Lie group written as a product of compact simple factors, then any bi-invariant second-order differential operator on G has the form*

$$D = c_0 \mathbf{1} - \sum_i c_i \Delta_{G_i},$$

where Δ_{G_i} is the Laplace–Beltrami operator on the factor G_i.

Remark A.3. *Bi-invariance is equivalent to $D \in Z(U(\mathfrak{g}))$. Linear terms vanish because there are no invariant vectors in the adjoint representation, and the quadratic part is proportional to the Killing form on each simple factor, giving one independent coefficient per factor. The Casimir element then acts as $-\Delta_G$ in the regular representation.*

A.4 Peter–Weyl Multiplicity and Beta Shifts

By Peter–Weyl decomposition, the effective refinement-limit edge representation space is

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*.$$

Entanglement traces over one factor give multiplicity d_R in p_R, but loops see both factors, so the effective multiplicity entering the calibration branch is

$$N_{\text{eff}}(R) = d_R\, p_R.$$

The one-loop beta shift from edge sectors is then

$$\Delta b_a = \sum_R p_R \, d_R \, T_a(R),$$

where $T_a(R)$ is the Dynkin index for gauge factor a. This is the exact mathematical point on the D10 lane where edge-sector heat-kernel weights feed the calibration branch. Matching to MSSM-like one-loop running remains a declared Phase II benchmark, not a promoted theorem-level MSSM spectrum claim.

At the branch unification diffusion parameter $t_U(P) \approx 1.64$ selected by the forward D10 solve, the benchmark shift is

$$\Delta b \approx (2.49, 4.38, 3.97) \qquad \text{vs MSSM} \qquad (2.50, 4.17, 4.00).$$

With the additional D10 calibration assumption of a fermionic-grading restriction to half-integer SU(2) sectors, this sharpens to

$$\Delta b \approx (2.50, 4.17, 3.97),$$

which matches the MSSM-style one-loop benchmark at the percent level under that declared calibration package. Equivalently, the benchmark ratio

$$\frac{\Delta b_3}{\Delta b_2} \approx 0.91$$

sits close to the MSSM comparison value 0.96.

References

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